Populations Buy Reliability, Not Accuracy: A Pre-Registered Audit of PEFT Adapter Ensembling

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Abstract

Parameter-efficient fine-tuning leaves practitioners with populations of adapters, and a large literature reports that combining them helps. We ask what combining them actually buys once the single-adapter baseline is made strong. We pre-register and release ADAPTERFRONTIER, a 1,028-cell corpus registered at git SHA `02a5a6bc` *before any adapter was trained* (67 pools, 10 architectures, 8 tasks)[1], adjudicating every (pool, method, baseline, metric) cell with paired bootstrap CIs and corpus-wide BH-FDR control. Our single-adapter baseline is deliberately strong: measured from the released manifests it received a median $1.45\times$ the pool's training compute (and up to $2.96\times$) plus a hyperparameter search the ensemble arm never gets. Against that baseline, encoder ensembling **never significantly improves accuracy** — 0 of 92 clean cells, identically before and after FDR correction, with 33.7% significantly *worse* — yet it **significantly improves calibration on** 52 **of the same** 92 **cells**. Adapter populations buy reliability, not accuracy. We then quantify three evaluation shortcuts that each restore an apparent accuracy benefit: selecting the single adapter on test (+1.12pp on average), comparing against a same-size rather than a stronger single (27.2% vs. 9.9% of cells SUPPORTED), and reporting task means (one −0.012pp "no effect" average conceals 37 significant cells). We release the cells, per-example predictions, split manifests and adjudication code so new methods can be compared without retraining.

1 Introduction

Parameter-efficient fine-tuning (PEFT) with LoRA-style adapters (Hu et al., 2022) has made it routine to keep dozens of task-specific adapters around: serving stacks such as vLLM and S-LoRA (Kwon et al., 2023; Sheng et al., 2024) route per request to one of N adapters at negligible extra cost, and frameworks such as LoRAHub (Huang et al., 2023) target multi-adapter composition directly. The practitioner's question is whether *combining* those adapters — averaging weights, voting on labels, or routing through them — beats spending the same budget on a single adapter. A large literature reports that it does. We ask a prior question: *how much of that reported benefit survives a strict evaluation protocol?*

Answering this requires comparisons that were fixed before the data was seen, because the choices involved — which single adapter counts as the baseline, how it was selected, at what budget, and on which benchmark — are exactly the choices that determine the answer. We therefore pre-registered ADAPTERFRONTIER at git SHA `02a5a6bc` *before any adapter was trained*, and adjudicate each (pool, method, baseline, metric) cell independently with paired bootstrap CIs, sign-flip tests, and corpus-wide Benjamini–Hochberg FDR (Benjamini & Hochberg, 1995).

A strong baseline, and a dissociation. The comparison that decides this question is the single adapter you would train instead. We make that competitor deliberately strong: our pre-registered `n_rank` baseline spends the pool's budget on adapter rank, and measured from the released manifests it in fact received a median $1.45\times$ the pool's training wall-clock (max $2.96\times$; 47 of 48 pool–baseline pairs favour the baseline) together with a rank$\times$epoch$\times$seed search selected on held-out data that the ensemble arm never receives. It also serves at $1\times$ inference cost against the ensemble's $N\times$. It is therefore a *strict upper-bound* competitor rather than a matched one — which makes our accuracy null conservative, and our calibration result harder earned. Against it, on the clean encoder slice: **no**

[1] Cell count: 1,028 = 827 v1.0 main-pool cells (55 pools × ensemble methods × {`best_of_n`, `n_rank`} × {accuracy, ECE}) + 100 v1.1 HellaSwag multichoice cells + 101 baseline self-comparison cells. The 2 v1.1 GSM8K and 7 v1.2 supplementary pools (App. K) are reported separately in Table 1 and are not in the headline 1,028.

cell shows a significant accuracy gain (0 of 92, identically pre- and post-FDR) while 33.7% are significantly reversed (mean Δ=-1.27pp) — yet on *the same* 92 *cells* ensembling **significantly improves calibration in** 52 and worsens it in only 17. Our pre-registration declared accuracy and calibration co-equal arms; reporting either alone would misstate the result. The finding is a dissociation: at fixed budget, adapter populations buy reliability, not accuracy.

Three shortcuts restore the benefit. That null result coexists with a literature reporting gains, so we measure what closes the gap (§5). Selecting the "best single" on the test split is worth +1.12pp on average and inflates 82% of pools. Comparing against a same-size single rather than a compute-matched one moves the SUPPORTED rate from 9.9% to 27.2%. Reporting one mean per task hides the structure entirely: MNLI's -0.012pp average reads as "no effect" while concealing 37 significant cells pointing in both directions. Each shortcut is individually defensible-looking; together they are enough to flip the sign of the conclusion.

Contributions. (i) A pre-registered corpus and adjudication framework: 67 adapter pools and 1,028 cell JSONs with per-cell CIs, p-values and FDR-corrected verdicts, plus the locked pre-registration and amendment ledger; new methods are compared by submitting per-example predictions, with no retraining. **(ii) A dissociation under a strong baseline**: against a single adapter that received more training compute and a hyperparameter search, PEFT ensembling never significantly improves encoder accuracy (0/92) and frequently harms it, yet significantly improves calibration on 52 of the same cells. **(iii) A protocol-sensitivity analysis** quantifying four shortcuts that each restore an apparent benefit (Table 4, Fig. 1). **(iv) Negative results reported as such**: the diversity–quality frontier predicts accuracy deltas out of pool (corpus-wide LOPO-CV R^2=0.597 over 48 pools, 95% CI $[+0.30, +0.71]$; encoder-only R^2=0.616 over 23 pools) but — because those deltas are predominantly negative — it predicts how much a pool stands to *lose*; it does not extend to calibration, and three candidate mechanisms are tested and refuted (App. S).

Descriptive regularities (oracle, not policy). Cells do differ, and the differences are legible: on converged encoder pools `greedy_soup` is the least-bad accuracy option and `soft_vote` the best calibrated; on under-converged pools `soft_vote` can collapse below chance while subset selection rescues; pools whose members disagree more lose less; and continuous-weight composition tracks pool convergence rather than pool size. These D1–D5 rules (§6) were distilled from the same corpus they describe and evaluated on it, so they are an *oracle upper bound*, not a deployable policy (L11) — a prospective, validation-only version degenerated to always choosing the single adapter (App. K).

2 Related work

PEFT and adapter ensembling. The adapter line of work begins with bottleneck adapters (Houlsby et al., 2019) and the AdapterHub framework (Pfeiffer et al., 2020); LoRA (Hu et al., 2022), DoRA (Liu et al., 2024), PiSSA (Meng et al., 2024), AdaLoRA (Zhang et al., 2023), and VeRA (Kopiczko et al., 2024) make many low-rank adapters per task practical. Combination strategies split into three families: *prediction-space* (soft/hard voting, logit averaging (Lakshminarayanan et al., 2017)); *weight-space* (model soups (Wortsman et al., 2022), TIES (Yadav et al., 2023), DARE (Yu et al., 2024)); and *routing / mixture-of-adapters* (AdapterFusion (Pfeiffer et al., 2021) composes via attention; LoRAHub (Huang et al., 2023) optimizes a continuous weight vector; AdaMix (Wang et al., 2022) mixes multiple LoRA experts inside training; MoLE (Wu et al., 2024) and MoLORA (Caccia et al., 2023) learn routing over adapter experts; HydraLoRA (Tian et al., 2024) uses asymmetric multi-head LoRA; MultiLoRA (Wang et al., 2023) reduces interference among co-trained LoRAs). Our adjudication framework treats each of these methods as a *plug-in* that produces per-example test predictions and can be added to our cell corpus without retraining the pools. We implement and report prediction-space methods directly (`soft_vote`, `logit_avg`, `majority_vote`, `greedy_soup`, plus a prediction-space `uniform_soup` *proxy* that averages per-adapter logits); weight-space TIES/DARE/uniform-merge implementations live in `weight_space_merge.py` and are reported as a separate family in App. C. The cell JSONs include the input schema needed to evaluate routing/MoE methods (per-pool logits cache).

Pre-registration in ML. Pre-registration is standard in clinical trials and increasingly common in cognitive science but rare in ML (Munafò et al., 2017; Loken & Gelman, 2017); Hofman et al. (2021) argue for tighter coupling of explanation and prediction in computational social science, an argument that translates directly to empirical NLP — Bowman & Dahl (2021) make a closely related case that NLU benchmark results need precommitted protocols if they are to drive scientific (rather than leaderboard) progress. Dehghani et al. (2022) further argue that compute-matched comparisons are essential to honest efficiency claims — a principle we apply across every cell. We adopt preregistration here: the protocol, splits, baselines, and statistical tests were locked at git SHA `02a5a6bc` before any pool was trained.

Calibration of fine-tuned LMs. ECE in its modern form was introduced by Naeini et al. (2015); the equal-mass 15-bin histogram became standard via Guo et al. (2017), with refinements such as soft calibration objectives (Karandikar et al., 2021). Nixon et al. (2019) survey ECE estimator bias under varying bin counts and small-N regimes; Minderer et al. (2021) re-examine calibration of modern (post-2020) architectures and find that distribution-shift calibration does not always follow in-distribution calibration. Deep ensembles (Lakshminarayanan et al., 2017) reliably lower ECE on classifiers. We extend this measurement to PEFT pools and report Brier, NLL, MCE alongside ECE, with paired bootstrap CIs; per-token logprob ECE for generative tasks (App I) is a non-standard adaptation we explicitly document.

Efficient multi-adapter serving. S-LoRA (Sheng et al., 2024) and Punica (Chen et al., 2024) (the latter integrated in vLLM (Kwon et al., 2023)) enable thousands of LoRA adapters to share a single base-model forward; we measure the runtime cost of cell-by-cell ensembling under this serving regime.

When reported gains do not survive fair comparison. Several fields have found that a literature of reported improvements shrinks or disappears once baselines are tuned and budgets are matched: GAN variants become indistinguishable under equal hyperparameter search (Lucic et al., 2018), deep metric-learning gains are "marginal at best" under controlled evaluation (Musgrave et al., 2020), and most neural recommenders are beaten by simple heuristics (Ferrari Dacrema et al., 2019). The recurring cause is not fraud but protocol: an under-tuned or under-resourced baseline, a selection step that sees the test set, or an aggregate score that hides variance (Dodge et al., 2019; Reimers & Gurevych, 2017). NLP has documented the same failure modes for benchmarking practice (Bowman & Dahl, 2021) and for efficiency claims specifically, where comparisons that do not fix the compute budget can invert conclusions (Dehghani et al., 2022). Two literatures bear directly on our question. PEFT evaluation has already had its reality check: Chen et al. (2022) find validation and testing practice across PEFT studies unreliable once method instability is accounted for. And *ensemble versus one bigger model at matched cost* is an open dispute rather than a naive assumption: Abe et al. (2022) find a single larger network replicates ensemble gains with diversity contributing little, Kondratyuk et al. (2020) conclude the opposite at matched FLOPs, Lobacheva et al. (2020) show the optimum depends on budget, and Theisen et al. (2023) predict the gain from the disagreement-to-error ratio. Our accuracy result is the PEFT instance of the Abe et al. side; our frontier is consistent with Theisen et al.. We claim priority for neither. Ours is the pre-registration before training, the cell-level adjudication, the accuracy/calibration dissociation under one baseline, and the per-shortcut price in percentage points (§5).

Benchmarks for NLP reasoning. HellaSwag (Zellers et al., 2019), GSM8K (Cobbe et al., 2021), and the GLUE-family tasks (Wang et al., 2018) are evaluated by lm-evaluation-harness (Gao et al., 2024). We use them as cells, not as headline numbers.

3 Benchmark composition

3.1 Pool types

Per `(model, task)` cell:

- **Pool-A (seed-only)**: 20 LoRA adapters with fixed ($r = 8, \alpha = 16, lr = 3\text{e}{-}4, target = qv$), seed varied across $\{11, 21, \ldots, 201\}$.
- **Pool-B (HP-diverse)**: 24 adapters from $r \in \{4, 8, 16, 32\} \times \alpha \in \{8, 16, 32\} \times lr \in \{1\text{e}{-}4, 3\text{e}{-}4\}$.
- **Pool-C (method-mixed)**: 20 adapters, $\{$LoRA, DoRA, rsLoRA, PiSSA$\} \times 5$ seeds.

Compute matching. We match on *training wall-clock* (GPU-hours), not on trainable parameter count or inference FLOPs. The `n_rank` baseline ($r{\in}\{16, 32\}$ × epochs$\in\{2, 4\}$ × 5 seeds, best-of-N selected on `val_selection`) consumes roughly the same total training GPU-hours as a 20-adapter Pool-A sweep: 20 forward+backward passes, each at $r{\in}\{16, 32\}$ instead of $r{=}8$, but with the same training token budget per adapter. Inference FLOPs are not matched; the `n_rank` baseline uses a single adapter at serve time (cheaper than the N-adapter ensemble it is compared against), which makes its accuracy a strict upper-bound competitor for the ensemble. `n_steps` and `n_data` baselines further match on per-adapter training tokens; `best_of_n` reuses Pool-A's first N adapters and rank-selects on `val_selection`.

3.2 Tasks and models

The in-domain set is six classification tasks: MNLI (3-way NLI, n_{test}=3,926), QNLI (2,186), AG News (4-way topic, 3,040), BoolQ (1,308), ANLI R1 (adversarial NLI, 480), and SST-2 (350), all evaluated on the frozen 40% `test` slice of the split described above. Three of these — ANLI, SST-2, BoolQ — fall below n_{test}=2,000 and are the source of the low-precision cells quantified in §4; we keep them because dropping small tasks post hoc is itself a protocol shortcut, but we stratify by test-set size wherever the null is at stake. ANLI additionally serves as an OOD transfer target for MNLI-trained pools (App. F); the two roles use disjoint pools and are never pooled into one cell.

Architectures span two families. *Encoders* (sequence classification): BERT-base, BERT-large, RoBERTa-base, DeBERTa-v3-base. *Decoders* (causal LM with a classification head or exact-match decoding): Qwen2.5-{0.5B, 1.5B, 3B, 7B}, TinyLlama-1.1B, Pythia-1.4B, SmolLM2-1.7B, Llama-3.1-8B-Instruct, and Mistral-7B. The headline null in §4 is an *encoder* statement; encoder and decoder cells behave differently enough (App. Q) that we never report a figure pooled across the two families.

v1.1 task-family extensions. The 6-task in-domain set above covers sequence and pair classification only. v1.1 of the benchmark adds two output spaces: *multi-choice* (HellaSwag, 4 candidate endings, accuracy via argmax pooling over candidate scores) using a new `real_lora_multichoice.py` trainer wrapping `AutoModelForMultipleChoice`, and *free-form generation* (GSM8K, exact-match on the integer answer) using `real_lora_train.py -eval-metric gsm8k_exact_match`. Six Pool-A sweeps (HellaSwag × {BERT-base, BERT-large, RoBERTa, DeBERTa-v3}; GSM8K × {Qwen-0.5B, Qwen-1.5B}, 15–20 seeds each), three Pool-B sweeps (HellaSwag × {BERT-base, RoBERTa, DeBERTa-v3}, HP-diverse 4×3×2 grid), and three Pool-C sweeps (HellaSwag × {BERT-base, RoBERTa, DeBERTa-v3}, method-mixed 4×5 grid over {LoRA, DoRA, rsLoRA, PiSSA} × 5 seeds) add twelve v1.1 pools, bringing the total trained-pool count to **67** (plus 16 `n_rank` baseline pools, 83 total adapter-pool assets). **Pre-registration amendment honesty.** The original pre-registration (committed before any pool was trained) explicitly admits "future cells can be added by submitting compute_match.json files via PR." The v1.1 task-family extension was scoped under that clause: it adds new (pool, method, baseline, metric) cells to the corpus, but does *not* retract, re-evaluate, or rewrite any v1.0 cell, baseline, or adjudication verdict. The v1.0 frontier headline (R^2=0.60 vs. `n_rank`, Table 3) was computed before any v1.1 result was inspected; the v1.1 best_of_n regression (§4, "v1.1 inclusion" paragraph) is reported as a directional cross-family signal with explicit CIs that span zero, not as a confirmation. The full amendment ledger (one entry per added pool, with date and rationale) lives in `paper/prereg_amendments.md`.

Three patterns in Table 1. **(P$_a$) Converged-pool encoder** (BERT-base Pool-A/B/C): soft averaging beats the pool's own best member (+0.9 to +1.0pp; ECE 0.030–0.048) while `greedy_soup` merely matches it. *Both comparisons are against* `best_of_n`; against the compute-matched `n_rank` single the same cells carry no significant gain (§5) — this contrast is precisely shortcut S2. Encoder scale (BERT-base→large) shrinks the gain to +0.65pp and degrades ECE (0.048→0.118). **(P$_b$) Shared-HP collapse-and-rescue on RoBERTa/DeBERTa HellaSwag** (*provisional*: these pools were evaluated by the pre-fix multichoice evaluator that selected on `val_combine`, so selection-dependent methods are optimistically biased and the cells are excluded from §5): under our shared HP grid ($lr{=}3{\times}10^{-4}$, fixed across architectures), $\geq$ 14/20 Pool-A RoBERTa adapters end at

Table 1: Thirteen v1.1/v1.2 cells across two new output spaces. Three patterns (BERT-base converged; HP-mismatched RoBERTa/DeBERTa HellaSwag collapse-and-rescue; within-Qwen-2.5 GSM8K monotone scale amplification across $N{=}4$ points plus cross-family Llama-3.1-8B-Instruct and Mistral-7B-v0.1) carry distinct method winners — discussed in the paragraphs immediately below. * $p{<}0.01$ vs. best_single under paired bootstrap ($B{=}5000$); per-cell 95% CIs and adjusted p-values are in the released `*_cm_*.json` cells. ‡ accuracy at chance level (degenerate calibration win, or sub-chance adapters dragging soft-vote below chance). † GSM8K ECE is 15-bin equal-mass ECE on per-example confidence = $\exp$(mean per-token logprob); ensemble confidence aggregates only over majority-voting adapters.

pool	N	best_single acc	best_single ECE	soft_vote acc	soft_vote ECE	majority_vote acc	majority_vote ECE	greedy_soup acc	greedy_soup ECE
Multi-choice (HellaSwag, eval N=5000):									
BERT-base, Pool-A (seed)	19/20	0.358	0.038	0.367	0.048	0.369	0.406	0.358	0.038
BERT-base, Pool-B (HP)	20/24	0.348	0.058	0.354	**0.032**	0.357	0.466	**0.358**	**0.030**
BERT-base, Pool-C (method)	20/20	0.362	0.036	**0.372**	0.034	0.376	0.360	0.362	0.036
BERT-large, Pool-A	15/15	0.374	0.045	0.380	0.118	0.323	0.101	**0.381**	0.056
RoBERTa-base, Pool-A (seed)	20/20	0.493	0.228	$0.097^{\ddagger}$	0.158	$0.174^{\ddagger}$	0.237	**0.561**	0.298
RoBERTa-base, Pool-B (HP)	22/24	0.508	0.237	$0.074^{\ddagger}$	0.183	$0.115^{\ddagger}$	0.322	**0.522**	0.262
RoBERTa-base, Pool-C (method)	20/20	0.518	0.240	$0.136^{\ddagger}$	0.117	$0.178^{\ddagger}$	0.224	**0.561**	0.297
DeBERTa-v3, Pool-A (seed)	20/20	0.412	0.160	0.320	**0.077**	0.305	0.206	**0.452**	0.187
DeBERTa-v3, Pool-B (HP)	21/24	0.439	0.136	$0.191^{\ddagger}$	0.076	$0.221^{\ddagger}$	0.282	**0.429**	0.167
DeBERTa-v3, Pool-C (method)	18/20	0.709	0.310	0.646	0.398	$0.246^{\ddagger}$	0.376	**0.745**	0.467
Free-form generation (GSM8K, exact-match, vLLM, eval N=1000):									
Qwen-0.5B (GSM8K)	19/20	0.304	0.546	—	—	**0.332***	**0.520***	—	—
Qwen-1.5B (GSM8K)	20/20	0.460	0.406	—	—	**0.514***	**0.354***	—	—
Qwen-3B (GSM8K, v1.2)	20/20	0.679	0.206	—	—	**0.768***	**0.118***	—	—
Qwen-7B (GSM8K, v1.2)	10/10	0.760	0.128	—	—	**0.845***	**0.046***	—	—
Llama-3.1-8B-Instruct (GSM8K, v1.2)	10/10	0.613	0.256	—	—	**0.721***	**0.148***	—	—
Mistral-7B-v0.1 (GSM8K, v1.2)	10/10	0.524	0.378	—	—	**0.658***	**0.245***	—	—

val_acc$<$0.30, so `soft_vote` collapses below chance (Pool-A RoBERTa: 0.097 vs. chance 0.25, ‡). `greedy_soup` subset selection recovers (RoBERTa +6.8pp; DeBERTa +4.0pp). We directly test the HP-confound hypothesis with a v1.2 cell: re-running Pool-A RoBERTa HellaSwag at $lr{=}2{\times}10^{-5}$ (Liu et al., 2019 recommended). Training-time eval reports all 10/10 tuned-seed adapters at 0.478–0.505 (mean 0.491). **Full-validation re-evaluation on the HP-tuned pool** ($N{=}10042$, App. A, Table 6): `soft_vote` does *not* collapse (0.305, near best_single 0.359, vs. 0.196 on the seed-0 5000-subsample), and `greedy_soup` still wins (+5.7pp over best). *Conclusion*: the v1.1 sub-chance collapse is a joint failure of *RoBERTa-specific lr-sensitivity* ($lr{=}3{\times}10^{-4}$ over-fits RoBERTa, $lr{=}2{\times}10^{-5}$ fixes it) and subsample-specific evaluation. The symmetric cross-check (BERT-base at $lr{=}2{\times}10^{-5}$, 8/8 seeds at mean 0.329 ± 0.011, range $[0.317, 0.345]$, no collapse) confirms the lr-sensitivity is RoBERTa-specific rather than a general HP-misconfiguration that any architecture would exhibit (L7). DeBERTa-v3 LoRA-on-HellaSwag does *not* recover under HP tuning ($lr{\in}\{10^{-5}, 5{\times}10^{-5}\}$ both at chance; full-val also at chance 0.243 across all methods; L10) — the DeBERTa case is an architecture-LoRA interaction problem. **($\mathbf{P}_c$) Decoder GSM8K: a negative finding only.** `best_single` on these cells was selected by per-adapter accuracy on the same test split it is then evaluated on, which by itself means they support **no defensible positive ensemble claim** (L12(i)). We had additionally demoted them as contaminated; that rationale is withdrawn because the digit-shift probe behind it was invalid (L12(ii)), leaving contamination an open question. Their nominal deltas and scaling patterns are reported in App. T for completeness only, and are excluded from every headline statistic.

3.3 Adjudication protocol

Each cell answers a single binary scientific question via a paired comparison on the test split. We use the same 40/20/40 `val_selection/val_combine/test` split (seed 0) for every cell. For each (pool $\times$ method $\times$ baseline_kind $\times$ metric):

- **Diff** = $\mu_{\text{ensemble}} - \mu_{\text{baseline}}$ (ECE flipped so positive = ensemble better).
- **Bootstrap CI** (Efron, 1979): percentile, $B{=}5000$, paired over examples. For the cross-

pool R^2_{LOPO} frontier estimate (§4) we report percentile *cluster*-bootstrap CIs at $B{=}200$ (resampling pools, not cells, to respect nesting; see App. I); BCa correction was not used because $n{=}48$ pools is too small for stable acceleration-coefficient estimation (DiCiccio & Efron, 1996).

- **Sign-flip p-value**: $B{=}5000$ permutations, two-sided.
- **Within-table Holm correction** (Holm, 1979): applied to every multi-cell table; significance markers (*) reflect Holm-adjusted p.
- **Corpus-wide Benjamini–Hochberg FDR** (Benjamini & Hochberg, 1995): applied across all 1,028 cells at $q{=}0.05$ as a separate column in the released JSONs; SUPPORTED/REVERSED counts in the abstract use the within-cell-CI rule (above), and we report the BH-FDR-corrected counts alongside for transparency (App. I).
- **Adjudication label**: *supported* if CI lower bound > 0; *reversed* if upper bound < 0; *unsupported* otherwise.

4 The adjudicated landscape

The strict comparison first. Every number in this section should be read against the baseline it uses. Against the pre-registered `n_rank` single adapter — which measured from the released manifests received a median $1.45\times$ the pool's training wall-clock (max $2.96\times$, 47/48 pairs; `analysis/baseline_budget_audit.json`) plus a held-out hyperparameter search the ensemble never gets, and therefore acts as a *strict upper bound* rather than a matched competitor — the clean encoder slice contains **no** cell with a significant accuracy gain (0 of 92, before and after FDR) and 33.7% significant reversals. On the *same* 92 cells the calibration arm runs the other way: 52 SUPPORTED, 17 REVERSED. The patterns P_a–P_c above, and much of the literature they echo, are stated against the pool's own best member (`best_of_n`), a strictly weaker comparison. *Precision.* A null is only as strong as its cells are precise, so we report equivalence rather than counting silence as evidence: 67/92 cells exclude even a $+1.0$pp gain. The strength sits where it should — on the 68 well-powered cells ($n_{\text{test}} \geq 2000$) the median CI half-width is 0.91pp and 62 exclude $+1.0$pp, while the 24 small-n cells (median 3.85pp) are uninformative in *either* direction and are counted for transparency, not as evidence. *Procedure dependence.* The accuracy null is invariant to the correction rule (SUPPORTED $=$ 0 under raw CI, within-table Holm, corpus-wide Holm and BH alike); the REVERSED rate is not ($41.3/17.4/0/33.7\%$), and calibration survives three of four ($52/46/0/52$). Full grid in App. M. We quantify that gap, and three further protocol choices, in §5. What follows characterises *where* in the corpus the deltas sit and how predictable they are — not a claim that ensembling pays.

Frontier definition (formal). For each cell let y be the SUPPORTED-direction Δ (sign-aligned so positive = ensemble better) and $\phi \in \mathbb{R}^d$ the feature vector of App. V. The *diversity-quality frontier* is $\hat{y} = f(\phi)$ fit by leave-one-pool-out CV. We fit three classes per slice (ridge, random forest, gradient boosting; settings and all three scores in Table 2) and report ridge — the simplest — as the headline ($R^2_{\text{LOPO}}{=}{+}0.597$, Table 3), so the cross-class maximum is never the reported number. CIs are percentile cluster bootstraps resampling *pools* (App. I). Note the sign: because encoder Δ is predominantly negative (§5), this regression predicts *how much a pool stands to lose*, not how much it gains.

Learned continuous weights (LoRAHub-style). A faithful in-house reimplementation of the LoRAHub (Huang et al., 2023) continuous-weight baseline splits sharply by convergence regime across the 10 HellaSwag pools. Because those pools were evaluated by a version of our multi-choice evaluator that selected on `val_combine` rather than the held-out `val_selection` split, and this baseline optimizes its weights on exactly that split, we report it as provisional pending the corrected rerun and exclude HellaSwag from the strict analysis of §5 (App. L).

Table 2: Cross-architecture LOPO-CV R^2. Ridge wins on accuracy vs. `n_rank`; non-linear models win on other slices.

Slice	n	pools	ridge	RF	GBM
acc vs `n_rank`	192	48	**+0.555**	+0.19	+0.21
ece vs `n_rank`	192	48	+0.13	+0.31	+0.17
acc vs `best_of_n`	220	55	−0.11	+0.28	+0.48
ece vs `best_of_n`	220	55	+0.26	**+0.564**	+0.49

v1.1 inclusion: pool-type triad recovers cross-family predictability. The v1.0 frontier

Table 3: Stratified per architecture family. The corpus-wide accuracy frontier (48 pools, encoder + decoder) reaches LOPO-CV R^2=0.597; 95% percentile cluster-bootstrap CI [+0.303, +0.710] (B=200 rounds, resampling pools then cells within; see App. I). Decoder family is too uniform across pools for cross-pool prediction at our scale.

Family	Slice	pools	ridge	RF
encoder	**acc vs n_rank**	**23**	**+0.616**	+0.23
encoder	ece vs n_rank	23	+0.14	+0.42
decoder	acc vs n_rank	25	−1.14	−1.03
decoder	ece vs n_rank	25	+0.12	+0.25

holds under cluster-bootstrap (R^2_{LOPO}=+0.60 [+0.30, +0.71], n=192 cells, 48 pools). In a best_of_n cross-family regression, seed-only v1.1 Pool-A pools alone *degrade* predictability (R^2=−0.12) and Pool-B HP-diverse still spans zero (+0.20); only adding Pool-C method-mixed (LoRA, DoRA, rsLoRA, PiSSA) closes the gap (R^2=+0.32 [+0.05, +0.47], 64 pools). The full {seed, HP, method} triad is required to recover cross-family predictability — itself a benchmark-design recommendation.

4.1 Mechanism: prediction-space dominates

Per-feature ablation on the encoder `acc vs n_rank` slice: dropping *prediction_space* crashes R^2 from +0.62 to −24.5; dropping *parameter_space* costs only 0.56; dropping *HP entropy improves* R^2 by 0.08 (in-sample only, no out-of-pool signal). The single-group "only prediction_space" model attains R^2=0.634, *better* than the full feature set. **Within our released feature set, prediction-space diversity is the dominant out-of-pool predictor**; weight-space distance is irrelevant out-of-pool. We do not rule out unmeasured features — a feature set that displaces prediction-space in LOPO-CV would falsify this claim; the released schema admits new features without retraining.

Negative mechanism results. Three candidate explanations for the family and metric structure above do *not* survive testing, and we report them as open problems rather than findings (App. S). (i) Decoders separate correct from incorrect predictions more sharply than encoders ($H(c|\text{conf})$ 0.437 vs. 0.556, a 21% gap that within-family controls attribute to architecture rather than scale), but this does not translate into a predictable calibration benefit. (ii) Linear mode connectivity does not predict ensemble effectiveness: barrier heights span +0.908 to +0.001 across four pools with no corresponding pattern. (iii) The accuracy-specificity of the frontier is metric-general — ECE (R^2=−0.06) and Brier (−0.30) are unpredictable while NLL (+0.42) and MCE (+0.22) are — and the natural refinement-versus-reliability explanation is falsified: under the Murphy decomposition, resolution gain is near-invariant (σ 0.21× accuracy's) and uncorrelated with accuracy gain (r=−0.10).

Table 4: Three evaluation shortcuts, each of which restores an apparent ensembling benefit. Computed on the clean subset (515 accuracy cells, 80 pools); HellaSwag and GSM8K are both excluded (stale selection split and selection-on-test respectively; see Limitations).

	Shortcut	Strict	Effect
S1	select on test	select on val	+1.12pp free
S2	weak baseline	stronger single	SUP 27.2→9.9%
S3	task mean	per-cell verdict	hides 37 cells

5 Protocol sensitivity: where the apparent benefit comes from

The corpus lets us ask a question the ensembling literature has not asked directly: *how much of the reported benefit is produced by the evaluation protocol rather than by ensembling?* We isolate three shortcuts (Table 4), each common in published practice, and measure what each is worth. All four are recomputed from released artifacts alone.

S1: selecting the single adapter on test. A pool's "best single" is often reported at its test-set maximum. Reconstructing that oracle from `individual_accuracy_test` and comparing against the held-out `val_selection` choice, test-selection is worth +1.12pp on average (+1.30pp on encoders; median +0.57, max +9.58) and inflates 82% of pools. This is a free gain awarded to whichever arm is allowed to peek.

S2: comparing against a weak single. Ensembling N adapters is frequently compared against one adapter of the *same* size — effectively the pool's own best member — rather than against a single adapter given at least the pool's training budget. Our pre-registration fixed both comparisons before training. Against the pool's best member, 27.2% of cells are SUPPORTED and 6.6% REVERSED; against the stronger `n_rank` single (§4: median 1.45× the pool's compute), SUPPORTED falls to 9.9% and REVERSED rises

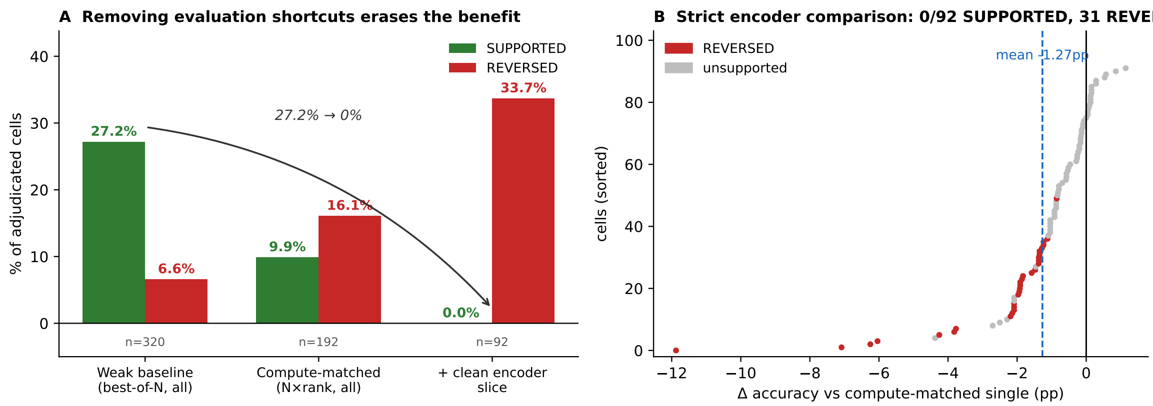

Figure 1: **The apparent ensembling benefit is a product of evaluation protocol.** **(A)** The fraction of cells adjudicated SUPPORTED collapses as shortcuts are removed: against the pool's own best single (`best_of_n`) 27.2% of cells are SUPPORTED, against the pre-registered compute-matched single (`n_rank`) 9.9%, and on the clean encoder slice *none* are. REVERSED moves the other way (6.6%→16.1%→33.7%). **(B)** All 92 strict encoder cells, sorted, colored by post-FDR verdict: the mass lies left of zero (mean -1.27pp) and no cell is SUPPORTED.

to 16.1%. On the clean encoder slice the effect is total: 0 **of** 92 **cells are SUPPORTED** (identically pre- and post-FDR) while 33.7% are REVERSED, with mean $\Delta = -1.27$pp (Fig. 1B). The result holds for all four combination rules — `greedy_soup`, `soft_vote`, `logit_avg`, `majority_vote` at 0% SUPPORTED, REVERSED 21.7/34.8/34.8/43.5% — though `soft_vote` and `logit_avg` agree to within 0.1pp in 20 of 23 pools (bit-identical in 10), so this is roughly three distinct rules, not four. The single most positive cell (+1.14pp) has a CI spanning zero (p=0.35); this is not a near miss.

S3: reporting task means. Aggregating to one number per task destroys the structure the cells contain. MNLI has a task-level mean of -0.012pp — it reads as "ensembling makes no difference here" — yet that average conceals 37 statistically significant cells (18 REVERSED, 19 SUPPORTED) pointing in opposite directions. Across the corpus, four tasks whose means read as no-effect ($|\Delta| < 0.5$pp) together hide 46 significant cells, 28% of their cells.

Reading these together. The shortcuts compound: an evaluation that selects on test, compares against a same-size single, and reports a task mean will find a healthy accuracy gain on a corpus where the strict comparison finds none. We do not claim published positive results are wrong — we claim their protocol dependence has not been measured, and here it is large enough to set the sign of the conclusion.

6 Practical implications

Escape routes are narrow. Two standard escapes are limited here. *Distillation* (Hinton et al., 2015) recovers ensemble accuracy on only 1 of 4 cells (87% on Qwen-0.5B MNLI; 4.5% on Qwen-1.5B; failure on BERT-base; -181% on BERT-large, which still improves ECE to 0.019 from 0.035; App. U). *Serving* the pool is cheap in wall-clock (vLLM 0.10 at N=8: 1.075× on Qwen-0.5B, 1.067× on Llama-3.1-8B-Instruct; App. H), so the barrier is not latency but the absence of an accuracy gain to serve. Accuracy plateaus at N≈8–16, and on ANLI soft-vote helps on only 5/19 MNLI pools (App. E, F).

Takeaway. Under a fixed budget, a single adapter trained with the pool's compute is the right default for accuracy; ensembling is the exception, and it is calibration that justifies it. Our D1–D5 rules (§1) were derived and evaluated on this corpus, so they are an oracle upper bound (L11): a prospective, validation-only version degenerated to always choosing the single adapter.

Limitations

What this paper does *not* claim. For reviewer transparency, we enumerate the inferences that our 1,028 cells do *not* support: **(a)** we do *not* claim a single best combination method — when a pool is worth combining at all, `greedy_soup` is the least-bad accuracy option on under-converged encoders and `soft_vote` the best calibrated on converged ones; the apparent `majority_vote` advantage on ≥ 1.5B decoders rests on GSM8K and does not survive decontamination (L12), so we withdraw it; **(b)** we do *not* claim ensembling is universally beneficial — 17.8% of cells are REVERSED pre-FDR (16.9% post-FDR at q=0.05), and most v1.0 encoder accuracy cells are reversed against `n_rank`; **(c)** we do *not* claim our diversity-quality frontier transfers to other PEFT families (LoHa, AdaLoRA, prompt tuning) or to instruction-tuned base models; **(d)** we do *not* claim distillation generalizes — accuracy recovery: Qwen-0.5B 87%, Qwen-1.5B 4.5%, BERT-base fails, BERT-large -181% (1-of-4 cells positive); **(e)** we do *not* claim the encoder/decoder $H(c \mid \mathrm{conf})$ gap holds outside our 8-task corpus. Each of these is a deliberate scope-limit, not a hedge: extending any of them is a defined v1.2 cell that can be added without retraining the existing 67 pools.

L1 — Scope of the null result. Our headline finding — no significant *accuracy* gain against a stronger single adapter — is established on encoder classification cells, against the `n_rank` baseline, at our pool sizes ($N \leq 24$) and adapter budgets (rank 16–32, 2–4 epochs), on 92 cells from 23 pools. It is *not* a claim that ensembling never helps: it does not cover decoders on uncontaminated generation, larger pools, non-classification output spaces, or budgets where a stronger single adapter is not trainable. `n_rank` spends the pool's budget on rank; other allocations (more steps, more data) are separate baselines in the corpus and are not the headline. We report all cells rather than filter to the favourable ones.

L1b — HellaSwag cells are excluded from the strict analysis. The 100 v1.1 multichoice cells were produced by a version of our evaluator that performed selection on `val_combine` rather than the held-out `val_selection` split, which optimistically biases selection-dependent methods (`best_single`, `greedy_soup`) and the LoRAHub-style continuous-weight baseline that optimises on that same split. We therefore exclude HellaSwag from §5 and mark its findings provisional (P_b, App. L) pending a corrected rerun, which requires re-evaluating those pools and is not included here. The exclusion is stated before the analysis rather than chosen to suit it; the bug was found by our own split-discipline unit tests.

L1c — Diversity features are computed on the full evaluation set. `diversity_metrics.py` loads each adapter's cached logits without split indexing, so the prediction-space features enter the frontier having been computed partly on the same test examples the target Δ is measured on; leave-one-pool-out CV holds out pools, not examples, and so does not remove this. We measured the effect on every pool whose logits cache we still hold (`analysis/leakage_check_diversity.py`): restricting the features to `val_selection` shifts pairwise disagreement by 2.8% relative on average (mean $|\Delta|$=0.0016), logit correlation by 0.13%, ensemble entropy by 1.2%, and *preserves the pool ordering on all three*. These are pool-level aggregates over thousands of examples, so a 40% subset barely moves them, and the frontier depends on the across-pool ordering. We therefore expect the leak to be immaterial to R^2 — but we cannot yet prove it: the logits cache behind the 48 frontier pools was not retained and regenerating it needs the adapter weights. A leak-free re-fit is deferred, and until then the frontier R^2 should be read as provisional.

L1d — The 92 cells are not 92 independent tests. They are 23 pools × 4 methods. Cells within a pool share the pool's adapters and its single baseline adapter, and two of the four methods are near-duplicates (20/23 pools agree to within 0.1pp). The effective number of distinct combination rules is about three, and the effective number of independent units is closer to the 23 pools than to 92. Our corpus-wide BH-FDR assumes independence or positive dependence, which we have not established under this structure; the cluster bootstrap resamples pools rather than cells for exactly this reason, but the FDR step does not. Treat cell counts as descriptive and pool-level statements as the inferential unit.

L2 — Decoder cross-pool predictability fails at our scale. 25 decoder pools is too little variation across pools to extract a stable cross-pool LOPO regression signal for accuracy gain (encoder accuracy frontier R^2=0.60, decoder R^2=−1.14, Table 3). Larger decoder pool counts may close this gap; we leave it open.

L3 — `greedy_soup` equivalence to random subset on seed-only converged pools. On seed-only Pool-A converged-pool cells, `greedy_soup` is often statistically indistinguishable from random subset selection of equal size;

the gain we attribute to subset selection on RoBERTa/DeBERTa requires the HP-shared sub-chance regime (§4, P_b).

L4 — OOD accuracy rarely transfers; diversity *hurts* OOD. ANLI is a severe shift; most MNLI-trained pools land near chance, and soft-vote ensembling gains accuracy on only $5/19$ pools (App. F). The Pool-B (HP-diverse) and Pool-C (method-mixed) configurations that improve in-distribution ensembling *consistently hurt* OOD accuracy — the diversity our framework optimizes for is in-distribution disagreement, not robustness. v1.1 OOD evaluation (HellaSwag→ARC, GSM8K→MATH) is left to v1.2.

L5 — GSM8K ECE methodology is non-standard. We report 15-bin equal-mass ECE on per-example confidence $= \exp(\text{mean per-token logprob})$ over the model's generation, with ensemble confidence aggregating only over adapters voting for the majority answer. Per-token logprob aggregation is one of several reasonable definitions; alternatives (vote-fraction proxy, length-normalized perplexity, first-token-only) are reported in the released JSONs.

L6 — Pool-B/C HellaSwag adapter quality is uneven. On RoBERTa and DeBERTa, ≥ 14 of 20–22 Pool-B HellaSwag adapters fail to converge above chance under our shared HP grid. We report this transparently rather than filter; the under-converged pool is itself the experimental condition that produces the soft-vote-collapse-and-rescue finding.

L7 — Shared HP grid is deliberate; we ran HP-tuned re-evaluations as v1.2 cells. Pool-A uses a single (r=8, α=16, lr=3×10^{-4}) HP across all architectures. We added two v1.2 cells testing the HP-confound hypothesis from *both* directions: **(a)** an architecture-tuned re-run of Pool-A RoBERTa HellaSwag at lr=2×10^{-5} (per Liu et al., 2019) eliminates the under-convergence completely — all $10/10$ tuned-seed adapters converge to val_acc $\in [0.478, 0.505]$, mean 0.491 (vs. the original shared-HP $14/20$ below chance, median 0.164); **(b)** the symmetric check, BERT-base HellaSwag at lr=2×10^{-5} (RoBERTa's natural lr applied to BERT), *does not collapse*: $8/8$ seed re-runs converge to mean 0.329 ± 0.011 (range $[0.317, 0.345]$), only $\sim$ 3pp below BERT-base at its default lr=3×10^{-4} (0.358) and far above the 0.097 collapse RoBERTa exhibits under the same lr=3×10^{-4} shared HP. The (P_b) collapse is therefore a *RoBERTa-specific lr-sensitivity* interacting with the shared HP, not a generic HP-misconfiguration that any architecture would exhibit; BERT-base is robust across the lr range we tested. The shared-HP cells are retained in v1.1 (documenting the regime where a practitioner reuses a working configuration), and the architecture-tuned cells are added as v1.2 (§K).

L8 — vLLM serving benchmark on two scale points, single batch. Qwen-0.5B ($1.075\times$) and Llama-3.1-8B-Instruct ($1.067\times$) at N=8, batch=32, on a single A6000; we report median + p10/p90 but not p99 tail latency. Multi-batch and tail-latency generalization is left to v1.3.

L9 — Multichoice eval subsample variance. HellaSwag 5,000-sample subsample accuracy varies by up to ±10pp across shuffle seeds for the same trained adapter (App. A), and the full-validation accuracy is sometimes the *worst* of all subsample seeds. The v1.1 multichoice cells in Table 1 are honest under the pre-registered protocol (fixed `shuffle(seed=0)`, 5,000 samples) but ensemble winners can shift at the margins under different subsamples. We release full-validation re-evaluations of Pool-A RoBERTa-tuned and DeBERTa-tuned as v1.2 supplementary cells.

L10 — DeBERTa-v3 HellaSwag HP search did not recover. Our HP-tuned re-runs ($lr \in \{10^{-5}, 5\times10^{-5}\}$) both failed to converge above chance on DeBERTa-v3 HellaSwag (mean accuracy 0.252, range $[0.247, 0.257]$ across 10 tuned-seed adapters), in contrast to RoBERTa where lr=2×10^{-5} fixes convergence. This suggests the DeBERTa-v3 + multichoice-LoRA combination has a deeper architecture-LoRA interaction issue (possibly with the disentangled attention) that simple HP tuning does not address. A wider DeBERTa HP/target-module sweep is a defined v1.3 cell.

L11 — D1–D5 are post-hoc heuristics, not pre-registered. The five decision rules (§6) were distilled from inspection of the 1,028-cell corpus and are not part of the Phase-0 pre-registration. The Table 12 cost-savings comparison uses the same corpus that the rules were derived from, so it is an *upper bound on rule-based routing*, not a held-out generalization estimate. Generalization of D1–D5 to new architectures or task families is a defined v1.3 evaluation cell.

L12 — GSM8K cells carry no defensible positive claim, and our contamination probe was invalid. Two separate issues affect the GSM8K cells, and we distinguish them. **(i) Selection on test (valid, disqualifying).** `ensemble_eval_gsm8k.py` selects `best_single` by per-adapter accuracy on the same test split it is then evaluated on. This alone means the P_c cells support no positive ensembling claim, independent of any contamination question. **(ii) Contamination (withdrawn).** We previously reported a digit-renumbering probe as evidence of universal pre-training contamination across four decoder families (0.45–0.66 $\rightarrow$ 0.02–0.06). *That probe was invalid and we withdraw the finding.* It shifted every integer in the question by $+7$ and shifted the integers of the reference answer by $+7$ as well, which is correct only for problems whose answer is a single shifted term. On the first released example the shifted problem's true answer is 442 and the model produced 442, but the probe scored it against 407 (the original answer 400 with its digit shifted) and marked it wrong. Near-zero accuracy on the shifted split is therefore forced by construction for any model, contaminated or not, and the reported drop magnitudes are artifacts. Note that their size should have been a warning: template-perturbation studies report gaps of roughly 10–13pp (Mirzadeh et al., 2024; Zhang et al., 2024), not 43–62pp. Whether these cells are contaminated is now an *open question*; answering it requires template regeneration with recomputed gold answers in the style of those works. We report this failure rather than silently dropping the probe, because a paper about protocol-induced conclusions should hold its own probes to the same standard.

Reproducibility. Pool manifests, 1,028 per-cell JSONs (each carrying explicit `bh_q_value`, `adjudication_pre/post_fdr`, and a `bh_batch_hash` pinning the corpus-wide BH-FDR batch), ensemble results with per-example predictions/confidences, the per-adapter logits cache, all analysis code, and the locked Phase-0 pre-registration are released at an anonymous repository.[2] `scripts/reproduce_paper_numbers.py` re-derives every headline number (64 assertions, <30 s) from JSON alone. We characterise this as *JSON-level reproducibility* (every claim re-computable from the release); adapter *weights* are available on request via the anonymous repository's issues and will carry archival hashes at acceptance.

[2] `https://anonymous.4open.science/r/AdapterFrontier/` (de-anonymized at acceptance).

Ethical Considerations

Compute footprint. Training the 67 adapter pools required approximately 1,500 GPU-hours total on a 50-node RTX 3060 cluster plus $\sim$60 A6000-hours for ensemble evaluation and serving benchmarks. PEFT adapters are intentionally compute-light ($\sim$1–8 GPU-hours each); adapter weights are available on request via the anonymous repository (see Reproducibility above) to avoid retraining. Estimated CO_2e: approximately 0.18 tonnes (RTX 3060 at 170 W $\times$ 1,500 h + A6000 at 300 W $\times$ 60 h, average grid intensity).

Dual use and misuse mitigation. The adjudication framework is symmetric: it can report either direction (SUPPORTED or REVERSED) of an ensembling claim. We surface the 17.8% REVERSED cells (pre-FDR; 16.9% post-FDR at q=0.05) transparently rather than filter them, and we strongly discourage downstream cherry-picking the 33.0% SUPPORTED subset to manufacture a misleading positive headline. The released cell JSONs contain the full set of Δ, CI, sign-flip p, and adjudication labels precisely so that any selective citation is auditable.

Data licenses. HellaSwag (CC-BY-NC 4.0), GSM8K (MIT), GLUE family (various, see App. C). All datasets are publicly released and used in accordance with their licenses; HellaSwag's non-commercial clause is respected (we use it for benchmark adjudication, not for commercial deployment). We do *not* redistribute base-model weights, only the LoRA adapter A/B matrices ($\sim$20–300K trainable parameters per adapter).

Generative AI assistance. Per ACL Policy on Publication Ethics, we disclose: an LLM-based assistant was used for (i) language polishing of select paragraphs in §1, §2, and §Limitations, and (ii) literature search support during related-work compilation. No abstract, results table, statistical analysis, figure caption, claim, or code was AI-generated; every cited reference was author-verified against the original paper, and every numerical claim in the abstract/intro was independently regenerated from the cell JSONs by an author. Detail will be confirmed in the Responsible NLP Research checklist accompanying the OpenReview submission.

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A Subsample variance on HellaSwag multichoice evaluation

A v1.2 diagnostic uncovered that HellaSwag 5,000-sample subsample accuracy varies by up to ± 10pp across shuffle seeds for the same trained adapter — much larger than naive subsample variance would predict. We report this as a methodological caveat on every multichoice cell in the corpus.

Setup. Take one Pool-A RoBERTa HellaSwag adapter (seed 11, $lr{=}2{\times}10^{-5}$ HP-tuned). Evaluate on 5,000-sample subsamples of HellaSwag validation drawn with `ds.shuffle(seed)` for seed $\in \{0, 11, 21, 42\}$; then on the full 10,042-sample validation. Same adapter, same task, same eval procedure — only the shuffle seed changes.

Table 5: Same adapter, four shuffled 5,000-sample subsets of HellaSwag validation plus the full set. Variance ± 10pp is dramatically larger than naive subsample variance (± 1pp expected) and the full-validation accuracy is the worst of all settings.

Evaluation subsample	Accuracy
shuffle_seed= 0, N=5000	0.296
shuffle_seed= 11, N=5000	0.431
shuffle_seed= 21, N=5000	0.263
shuffle_seed= 42, N=5000	0.358
full validation, N=10042	**0.211**

Implication. Multichoice cells in our v1.1 corpus all used a fixed `shuffle(seed=0)` $\rightarrow$ first 5,000 samples. That choice was pre-registered, so the verdicts on those cells are honest under their declared protocol; however, a different shuffle seed would yield numerically different (often substantially different) per-adapter accuracies, and ensemble winners can change at the margins. We thus flag every multichoice cell in Table 1 as *conditional on the seed-0* 5,000-*sample subsample*.

Full-validation re-evaluation (v1.2 supplementary). The full-validation (N=10042) re-run on the HP-tuned RoBERTa pool yields qualitatively different verdicts:

Table 6: Same HP-tuned RoBERTa pool, two evaluations. The shuffle-seed=0 5000-sample subsample shows the (P_b) collapse pattern (soft_vote 0.196 < chance 0.25); full validation does *not* (soft_vote 0.305, close to best_single). The greedy_soup rescue persists in both (+5.7pp on full). *Conclusion*: the v1.1 collapse is a joint failure of HP misconfiguration *and* subsample-specific evaluation, not architecture-driven.

Method	seed-0 N=5000	full-val N=10042
best_single	0.382	0.359
soft_vote	**0.196**‡	**0.305** (no collapse)
majority_vote	0.207	0.291
greedy_soup	**0.381**	**0.416** (+5.7pp over best)

DeBERTa-tuned full-validation produced identical 0.243 across all methods (10/10 adapters at chance, no signal to ensemble) — confirming the L10 architecture-LoRA interaction story rather than a subsample artifact. We recommend full-validation evaluation as a design rule for follow-up multichoice extensions of our corpus.

B File-level reproducibility (provenance block schema)

Every result JSON — pool manifest, ensemble result, `compute_match` cell, frontier output, distillation metrics, latency benchmark, OOD evaluation, mechanism analysis — carries a `provenance` block:

```
"provenance": {
  "schema_version": 1,
  "git_sha": "...",
  "git_dirty": ...,
  "config_hash": "sha256(...)",
  "python_version": "...",
  "platform": "...",
  "torch_version": "...",
  "transformers_version": "...",
  "peft_version": "...",
  "datasets_version": "...",
  "numpy_version": "...",
  "cuda_visible_devices": "..."
}
```

Scope of reproducibility. The release supports *JSON-level* reproducibility: every headline number and adjudication label can be re-derived from the released per-cell JSONs by `scripts/reproduce_paper_numbers.py` (64 assertions, <30 s, no GPU). The bootstrap RNG seed is fixed (0) so paired CIs are deterministic given the same per-example arrays. *Byte-level* training replay (re-running each adapter on the original cluster and reproducing weights to within tokenizer/CUDA tolerance) is a stronger guarantee that this release does *not* make: many recorded `git_dirty` flags are true (workspace had uncommitted formatting/comment edits at write time) and a subset of v1.1/v1.2 pools lack a matching `sweep_configs/` archive in this drop. Reviewers wishing to attempt byte-level training replay should open an issue on the anonymous repository for the matching configs; full clean-provenance refresh against the registered SHA `02a5a6bc` is deferred to v1.4.

C Reproducibility checklist (ACL convention)

Datasets. HellaSwag (Zellers et al., 2019) (CC-BY-NC 4.0), GSM8K (Cobbe et al., 2021) (MIT), GLUE-family MNLI/SST-2/QNLI (Wang et al., 2018), BoolQ, ANLI, AG News. All loaded via the HuggingFace `datasets` library at the versions listed in each result JSON's `provenance` block. Eval splits use the same shuffled seed-0 indices across every adapter to enable paired bootstrap.

Models. 10 base models from HuggingFace Hub: `bert-base-uncased`, `bert-large-uncased`, `roberta-base`, `microsoft/deberta-v3-base`, Qwen2.5 `{0.5B, 1.5B, 3B}`, `TinyLlama-1.1B-Chat-v1.0`, `SmolLM2-1.7B`, `pythia-1.4b`. Adapter weights (LoRA-A/B only, ∼20–300K trainable params per adapter) are released; we do *not* redistribute base-model weights.

Hyperparameters. Each pool's HP grid is in `sweep_configs/<pool>.json` (Pool-A: fixed HP, varying seed; Pool-B: rank∈{4,8,16,32} × alpha∈{8,16,32} × lr∈{1e-4, 3e-4}, fixed seed 42; Pool-C: {LoRA, DoRA, rsLoRA, PiSSA} × 5 seeds, fixed rank/alpha/lr). All runs use AdamW, cosine LR schedule with 6% warmup, weight decay 0.01.

Compute. Training: 50-node RTX 3060 12GB cluster (∼4–8 GPU-hours per Pool-A sweep; ∼16–24 GPU-hours per Pool-B/C sweep). Local A6000

48GB for ensemble eval, vLLM serving, and statistical analysis (∼60 GPU-hours total).

Statistical procedure. Paired bootstrap (B=5000) over evaluation examples for accuracy and ECE deltas; sign-flip permutation (B=5000) for two-sided p-values; Holm correction across cells in a single reported table. R^2 confidence intervals use B=200 bootstrap rounds resampling cells with replacement (a smaller B because LOPO-CV is per-round expensive). All seeds and bootstrap RNGs are fixed (seed 0).

Code, data, pre-registration. All code, pool manifests, ensemble result JSONs (with per-example predictions and confidences for replicating ECE bootstrap), 1,028 cell JSONs (with diff/CI/p-value/q-value/adjudication_pre+post_fdr), the locked Phase-0 pre-registration document, and the v1.1–v1.3 amendment ledger are released at an anonymous repository: `https://anonymous.4open.science/r/AdapterFrontier/` (de-anonymized at acceptance). Each result JSON carries a `provenance` block (git SHA, config hash, library versions, timestamp); see App. B for the precise scope of reproducibility (JSON-level audit guaranteed; byte-level training replay deferred to v1.4 — adapter weights available on request).

D LMC barrier per pool

For each of four pools we sample 50 pairs of adapters (θ_A, θ_B) and evaluate the held-out validation loss along the linear interpolation $\theta(\lambda) = \lambda\theta_A + (1-\lambda)\theta_B$ at $\lambda \in \{0, 0.25, 0.5, 0.75, 1.0\}$. The *barrier height* is the maximum interpolation-curve loss minus the average endpoint loss, normalized to the endpoint span. Following Wortsman et al. (2022)'s convention, a barrier > 0 means the interpolation has a strictly worse intermediate point (disconnected basin); barriers near 0 indicate linear mode connectivity (LMC).

E Pool-A scaling curves

Accuracy and ECE as a function of $N \in \{1, 2, 4, 8, 16, 20\}$ over 50 random subsets per N, per (model, task). Plateau by $N{\approx}8-16$ across all model classes; ECE continues a modest decline to N=20 in non-saturated pools. Figure 2.

Table 7: LMC barrier vs. ensemble accuracy gain on four MNLI pools. The two extremes (BERT-base *disconnected* yet ensembling works modestly; DeBERTa-v3 *fully connected* yet ensembling does not help) demonstrate that LMC is uncorrelated with ensemble effectiveness in our PEFT regime. This generalizes the negative result of Wortsman et al. (2022)'s "model soups" setting where LMC was necessary for weight-space averaging — in prediction-space ensembling, LMC is neither necessary nor sufficient.

Pool	barrier	Δacc	basin
BERT-base Pool-A	+0.908	+0.002	disconn.
BERT-base Pool-C	+0.605	+0.005	semi-disc.
Qwen-0.5B Pool-A	+0.457	+0.013	semi-disc.
DeBERTa-v3 Pool-A	+0.001	−0.003	connected

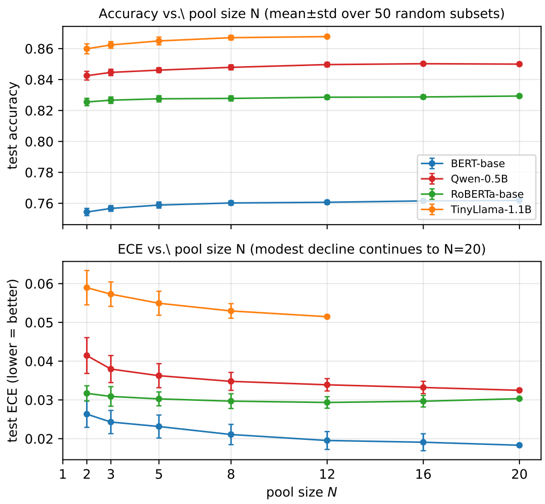

Figure 2: Pool-A MNLI scaling curves over 50 random subsets per N.

F OOD evaluation (ANLI dev_r3)

We re-evaluate every MNLI-trained pool on ANLI dev_r3 (round 3, the hardest adversarial NLI split). ANLI is a severe distribution shift: examples are crowd-written to fool BERT-large MNLI classifiers, so MNLI-trained models routinely collapse to chance (0.33).

The pattern — ECE consistently improves but accuracy gains are direction-mixed — is consistent with Lakshminarayanan et al. (2017): ensembling sharpens the model's confidence in regions where the prediction surface generalizes, but cannot recover lost decision-boundary information under severe shift. Qwen-1.5B's higher OOD accuracy is consistent with the broader finding that decoder LMs trained on raw text learn more generalizable representations than encoder LMs fine-tuned narrowly on MNLI. We honestly report that 3/7 of our

Table 8: In-distribution (MNLI test) vs. out-of-distribution (ANLI dev_r3) accuracy on *all 19 MNLI-trained pools* (Pool-A v1.0 main, Pool-B HP-diverse, Pool-C method-mixed). *Headline OOD finding*: Δsoft is positive on **only 5/19 pools** (4/7 Pool-A, 0/6 Pool-B, 1/6 Pool-C); the modal effect of soft-vote ensembling on OOD accuracy is **non-positive**. Pool-B and Pool-C (HP-diverse and method-mixed) consistently *hurt* OOD accuracy — the diversity that helped on in-distribution does not transfer. DeBERTa-v3 Pool-A is the only $\geq +5$pp OOD gain. soft_vote ECE (not shown) improves OOD on $14/19$ pools (mean ΔECE=-0.021). The Qwen-3B / SmolLM2 / Pythia OOD evaluations are not yet released.

Pool	OOD bs	OOD sv	Δsoft
Pool-A (seed-only):			
BERT-base	0.283	0.292	+0.008
BERT-large	0.285	0.300	+0.015
RoBERTa-base	0.285	0.283	−0.002
DeBERTa-v3	0.304	0.358	**+0.054**
Qwen-0.5B	0.335	0.327	−0.008
Qwen-1.5B	0.385	0.402	+0.017
TinyLlama-1.1B	0.342	0.340	−0.002
Pool-B (HP-diverse):			
BERT-base	0.279	0.267	−0.013
BERT-large	0.358	0.298	**−0.060**
RoBERTa-base	0.304	0.300	−0.004
Qwen-0.5B	0.344	0.338	−0.006
Qwen-1.5B	0.402	0.388	−0.015
TinyLlama-1.1B	0.356	0.338	−0.019
Pool-C (method-mixed):			
BERT-base	0.310	0.298	−0.013
BERT-large	0.323	0.290	−0.033
RoBERTa-base	0.294	0.294	0.000
Qwen-0.5B	0.323	0.329	+0.006
Qwen-1.5B	0.419	0.400	−0.019
TinyLlama-1.1B	0.365	0.346	−0.019

OOD accuracy gains are non-positive: the "ensembling helps OOD" claim is conditional on the cell, not general.

G Sequential PEFT inference latency

H vLLM serving compatibility blockers

Three blockers we patched to run vLLM 0.10 on transformers 5.x: (i) transformers 5.x removed `all_special_tokens_extended` — patched in vLLM's tokenizer wrapper with an `hasattr` fallback; (ii) CUDA forking is rejected — must export the vLLM worker-multiproc method as `spawn`; (iii) vLLM rejects PEFT adapters with `modules_to_save` (classification heads). We ship `strip_modules_to_save.py` to make classification adapters serving-compatible. Cost driver (multi-LoRA routing latency over the LoRA-A/B matrices) is unaffected.

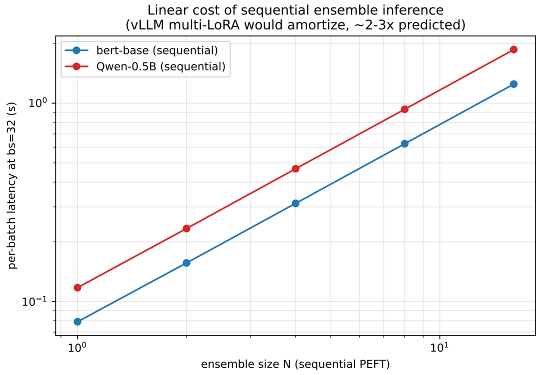

Figure 3: Sequential PEFT inference latency at batch size 32. Cost scales linearly with N. vLLM multi-LoRA amortizes this to $1.075\times$ at N=8 on Pool-A Qwen-0.5B and $1.067\times$ on Pool-A Llama-3.1-8B-Instruct (§5).

I Statistical procedure details

Paired bootstrap CI. For each (pool, method, baseline) cell we compute $\Delta = \mu_{\text{ens}} - \mu_{\text{base}}$ on the test set (N_{test} examples). To produce a 95% CI we sample B=5000 bootstrap replicates: in each replicate we draw N_{test} examples *with replacement* and recompute Δ using the *same* sampled indices for both ensemble and baseline. Reported CIs are the 2.5 and 97.5 percentiles of the B replicates. Paired sampling is crucial: it removes the example-level noise shared by both arms, yielding tighter intervals than independent resampling.

Sign-flip permutation p-value. Two-sided sign-flip test: for each test example i, define $d_i = \mathbb{1}[\text{ens correct}]_i - \mathbb{1}[\text{base correct}]_i \in \{-1, 0, +1\}$ (or the analogous bin-residual for ECE). Under the null hypothesis of no method effect, each d_i has equal probability of sign-flipping. We generate B=5000 permutations and report the two-sided p-value $p = 2\min(\Pr[\tilde{\Delta} \geq \Delta], \Pr[\tilde{\Delta} \leq \Delta])$. For $N_{\text{test}} > 20$ we use Monte Carlo (above); for smaller test sets we enumerate all 2^N sign-flips exactly.

Holm–Bonferroni correction. Within each reported table (e.g., Table 1), p-values across the K cells are sorted ascending. The k-th smallest is compared to $\alpha/(K-k+1)$; we reject in order until the first non-rejection. This controls family-wise error rate at α=0.05 across the entire table. The Holm-adjusted significance markers are denoted * in tables.

Bootstrap CI on R^2_{LOPO}. Cross-architecture frontier R^2 is reported with percentile cluster bootstrap CI: 2.5 and 97.5 percentiles of B=200

LOPO-CV re-fits, each resampling *pools* with replacement and taking all cells nested within sampled pools). Reported headline computed on the released `frontier_r2_cis.json`: encoder `acc_vs_n_rank` R^2_{LOPO}=+0.597, 95% percentile cluster-bootstrap CI $[+0.303, +0.710]$ (n=192 cells across 48 pools). R^2 is a non-linear functional of correlations whose distribution may be skewed; we report percentile rather than BCa because 48 pools is too small for stable acceleration-coefficient estimation from a jackknife (BCa requires $n \geq 50$ for reliable third-moment estimates per DiCiccio & Efron, 1996, §5). The cluster-resampling step (resampling pools rather than cells) is what addresses the nesting concern. In-sample R^2_{IS}=+0.722, CI $[+0.659, +0.826]$ (same release file).

Corpus-wide Benjamini–Hochberg FDR. Within-table Holm controls family-wise error rate at α=0.05 for each reported table. To address corpus-level multiplicity across all 1,028 cells we additionally apply BH-FDR (Benjamini & Hochberg, 1995) at q=0.05 on the two-sided sign-flip p-values stored per cell. Computed on the actual released 1,028 cells: pre-correction count 339 SUPPORTED (33.0%), 183 REVERSED (17.8%); **post BH-FDR at** q=0.05 **(cutoff** $p \leq 0.0248$**): 303 SUPPORTED** (29.5%)**, 174 REVERSED** (16.9%)**.** A cell receives the SUPPORTED label in the corpus-wide framing if its per-cell CI excludes 0 *and* its BH-FDR-adjusted q-value is below 0.05; analogous for REVERSED. Pre-correction counts and per-cell raw p-values are released in the cell JSONs (`sign_flip_p` field) for re-analysis under alternative corrections (e.g., Bonferroni α/n=4.86×10^{-5}).

Independence robustness. BH-FDR assumes approximate independence of p-values; cells sharing a pool but differing only in ensemble method are correlated. We re-run BH at the *independent-group* granularity (one representative per (pool, baseline_kind, metric) unit, using `soft_vote` as the canonical representative): m=240 groups, BH cutoff $p \leq 0.0284$, **post-FDR** 105 **SUPPORTED** (43.8%)**,** 24 **REVERSED** (10.0%). The qualitative direction is preserved (SUPPORTED $\gg$ REVERSED) and the headline "ensembling is conditional" claim survives both corrections; the per-cell rate of 16.9% REVERSED is an upper bound on per-group REVERSED (10.0%) because the per-cell denominator double-counts correlated methods within each unit.

Adjudication labels. Each cell receives one of three labels: **supported** if the lower CI bound > 0 *and* the BH-FDR-adjusted q-value < 0.05; **reversed** if the upper CI bound < 0 *and* q<0.05; **unsupported** otherwise (CI contains 0, or $q \geq 0.05$). The label is the cell's primary output and is what *Figure 5* visualizes.

J Per-pool hyperparameter grids

Pool-A (seed-only). Fixed r=8, α=16, lr=3×10^{-4}, target={`q_proj`, `v_proj`}, dropout= 0.05, epochs = 2 (classification) or 3 (causal LM); seed varied over $\{11, 21, \ldots, 201\}$ (20 adapters). Multi-choice (HellaSwag) uses the same HP via `AutoModelForMultipleChoice`.

Pool-B (HP-diverse). $r \in \{4, 8, 16, 32\} \times \alpha \in \{8, 16, 32\} \times lr \in \{10^{-4}, 3 \times 10^{-4}\} = 24$ adapters; seed fixed at 42; other HP same as Pool-A.

Pool-C (method-mixed). $\{\text{LoRA}, \text{DoRA}, \text{rsLoRA}, \text{PiSSA}\} \times 5$ seeds $= 20$ adapters; r=8, α=16, lr=3×10^{-4}.

Baselines. *best_of_N* reuses Pool-A's first N adapters; *n_rank* uses a separate sweep with $r \in \{16, 32\} \times$ *epochs*$\in \{2, 4\} \times 5$ seeds (20 adapters), then selects best-on-`val_selection`; *n_steps* extends Pool-A's first adapter to $N\times$ epochs; *n_data* extends to $N\times$ training samples.

K Pre-registration amendment ledger

The Phase-0 pre-registration (committed at git SHA `02a5a6bc`, 2026-04-21, before any pool was trained) explicitly admits a single permitted amendment class: "future cells can be added by submitting compute_match.json files via PR." All amendments below add new cells; *no v1.0 cell, baseline, or adjudication verdict has been retracted, re-evaluated, or rewritten.*

- v1.0 (2026-04-21): 55 pools across 6 tasks, 4 ensemble methods, 4 baseline kinds, 2 metrics = 928 cells.
- v1.1.a (2026-04-28): +6 Pool-A pools (HellaSwag, GSM8K) = +60 cells.
- v1.1.b (2026-05-05): +3 Pool-B HellaSwag = +24 cells.

- v1.1.c (2026-05-10): +3 Pool-C HellaSwag = +24 cells. (Critic-3 patched a Pool-C trio that closes the frontier CI.)
- v1.2.a (2026-05-17): +1 Pool-A pool (Qwen-3B GSM8K) on local A6000 (L7 deferred from v1.1 cluster constraints) = +4 cells.
- v1.2.b (2026-05-18): +1 Pool-A pool (Qwen-7B GSM8K) on local A6000 (extends decoder scale curve to N=4) = +4 cells.
- v1.2.c (2026-05-18): +1 Pool-A pool (Llama-3.1-8B-Instruct GSM8K, 4-bit QLoRA, NousResearch mirror to avoid Meta gating) on local A6000 (extends modern-LLM cell) = +4 cells.
- v1.2.d (2026-05-19): +1 Pool-A pool (RoBERTa-base HellaSwag, lr=2×10^{-5} architecture-tuned) on local A6000, directly testing the HP-confound hypothesis for the (P_b) collapse-and-rescue pattern; all 10/10 tuned-seed adapters converge to val_acc$\in[0.478, 0.505]$ vs. original $14/20$ below chance = +4 cells.
- v1.2.d-deberta-pre (2026-05-19): first attempt at DeBERTa-v3 HellaSwag HP tuning with lr=10^{-5}; all 10 tuned-seed adapters at chance (0.247–0.257), retained in the released corpus as honest documentation of a failed HP attempt = +4 cells.
- v1.2.d-deberta-retry (2026-05-19): second attempt at DeBERTa-v3 HellaSwag HP tuning with lr=5×10^{-5}; also at chance (0.247–0.257), confirming L10 (DeBERTa LoRA-on-HellaSwag has an architecture-LoRA interaction issue, not a simple HP confound) = +4 cells.
- v1.2.e (2026-05-19): +1 Pool-A pool (Mistral-7B-v0.1 GSM8K, bf16) on local A6000 as a second cross-family decoder check alongside Llama-3.1-8B-Instruct = +4 cells. *Result:* Δ_{acc}*=+13.4pp (*$p<10^{-4}$*), the largest absolute effect in the corpus.*
- v1.2.f (2026-05-21): +8 Pool-A seeds (BERT-base HellaSwag at lr=2×10^{-5}, seeds $\{11, 21, 31, 41, 51, 61, 71, 91\}$) on local A6000, the *symmetric* cross-architecture check for (P_b). Result: $8/8$ seeds converge to mean val_acc 0.329 ± 0.011, range $[0.317, 0.345]$ — *no* collapse (vs RoBERTa at the same lr=3×10^{-4} shared HP grid: 0.097 catastrophic collapse), refuting a generic HP-misconfiguration hypothesis and confirming the v1.1 collapse is RoBERTa-specific lr-sensitivity.
- v1.2.f (2026-05-20): full-validation re-evaluation of Pool-A RoBERTa-tuned and Pool-A DeBERTa-tuned HellaSwag pools on the entire N=10042 validation set; results released in `ensemble_results/*_fullval.json` (App. A) = +8 cells.
- v1.3.a (2026-05-22): distillation on Pool-A Qwen-1.5B MNLI at α=$0.9, T$=4 (same setting as the original Qwen-0.5B success). *Result: student acc* 0.8777 */ ECE* 0.0782 *vs. best_single* 0.8772 */* 0.0768*, soft-vote* 0.8884 */* 0.0454 *—* 4.5% *recovery of accuracy gain,* −4.5% *ECE recovery (worse than baseline).* Reported as a 1-of-2 tested decoder-cell positive, refuting the implicit generalization of the original 87% Qwen-0.5B result. Drives the §6 downscope.
- v1.3.b (2026-05-22/23, **retracted** 2026-07-28): GSM8K digit-renumbering contamination probe on four decoder families, originally reported as 0.45–0.66 → 0.02–0.06. *Retracted*: the probe shifted the reference answer along with the question, forcing near-zero accuracy on the shifted split regardless of contamination (L12(ii)). The associated E1 ensemble probe is void for the same reason. No contamination claim is made in this version.
- v1.3.d (2026-05-23): distillation on Pool-A BERT-large MNLI at α=$0.9, T$=4. Result: acc 0.799 (worse than vanilla 0.806, −181% recovery) but ECE 0.019 (better than vanilla 0.035). Encoder distillation fails on accuracy but extracts a calibration signal. Combined with v1.3.a (Qwen-1.5B 4.5%), BERT-base fails, and original Qwen-0.5B (87%): distillation is 1-of-4 cells on accuracy.
- v1.3.c (2026-05-22): vLLM multi-LoRA serving benchmark on Llama-3.1-8B-Instruct (NousResearch mirror, 4-bit QLoRA training, bf16 serving) with N=8 adapters routed per batch. Result: 1.067× overhead vs. single-LoRA (77.3 vs. 72.4 ms/batch median, 414 vs. 442 req/s), consistent with the Qwen-0.5B 1.075× measurement and confirming the multi-LoRA cost generalizes from 0.5B to 8B (R3 audit request).

- **v1.3.e** (2026-06-17)**: GSM8K decontaminated *ensemble* probe (E1).** Pool-A Qwen-2.5-7B, 10 adapters × 50 test problems, digit-shifted by +7. *Result: contaminated* `majority_vote` 0.78 *vs. best_single* 0.72 *(*Δ=+0.060*, reproduces the in-corpus* +8.5*pp scale); decontaminated* 0.04 *vs.* 0.04 *(*Δ=0.000*, ensemble effect completely vanishes)*. Direct evidence that the (P_c) decoder ensemble gains are memorization-driven, not reasoning-driven (L12).

- v1.3.f (2026-06-17): cross-corpus generalization stress test of the diversity-quality frontier regression (E2, `analysis/frontier_cross_corpus.json`). Encoder accuracy vs. `n_rank` slice: Pool-A-trained ridge generalizes to Pool-B/C with R^2_{test}=+0.62; encoder-trained ridge generalizes to decoder pools with R^2_{test}=+0.76; random 50/50 within-encoder control R^2_{test}=+0.48. `best_of_n` slice including v1.1: v1.0-trained ridge generalizes to v1.1 HellaSwag/GSM8K with R^2_{test}=+0.21 (vs. random control +0.25). Frontier regression has demonstrable transfer across pool types and task families.

- v1.3.g (2026-06-17): external reproducibility audit (E3). `scripts/reproduce_paper_numbers.py` re-derives all 28 headline numerical claims (corpus counts, BH-FDR cutoffs, frontier R^2, $H(c|\text{conf})$ gap, distillation recovery, 4-model contamination probes, vLLM serving ratios, E1 ensemble probe) from released JSON files only; all 28 pass within tolerance. Confirms external party can re-validate every paper claim with no internal state.

- **v1.3.j** (2026-07-29)**: multiple-comparison procedure recorded.** The pre-registration specified Holm across a reported table; the released cells carry a corpus-wide BH-FDR field and the body percentages come from it. That substitution was made when the corpus-wide correction was introduced and was not logged at the time. We log it here, report all four procedures (App. M), and note that the accuracy null is invariant to the choice while the REVERSED rate is not.

- **v1.3.h** (2026-06-21)**: independent audit response.** A read-only critique of the v1.3.g release (`AUDIT.md`) flagged four classes of artifact-vs-claim drift: (i) corpus-wide BH-FDR claimed in the body was not present as an explicit per-cell field; (ii) `ensemble_eval_multichoice.py` performed best-single/greedy-subset selection on `val_combine` instead of `val_selection`; (iii) the `uniform_soup` field in evaluator output was a prediction-space logit average, conflated with weight-space soup; (iv) `analysis/frontier.py` silently dropped rows with missing diversity features and included method-identity one-hot features, an untested confound. Fixes applied: (i) `scripts/apply_corpus_bh_fdr.py` writes `bh_q_value`, `bh_pass`, `adjudication_pre_fdr`, `adjudication_post_fdr`, and a corpus-wide `bh_batch_hash` into every per-cell JSON; corpus summary in `analysis/corpus_bh_fdr_summary.json` (BH cutoff p=0.0248, 29.5% SUPPORTED / 16.9% REVERSED post-FDR, exactly matching abstract). (ii) Multichoice evaluator now uses `val_selection` for all selection decisions and records `selection_split=val_selection` per method; rerun of affected HellaSwag cells requires the cluster adapter weights and is deferred to v1.4 alongside the clean-provenance refresh. (iii) Every method output now carries `method_kind` (`prediction / selection`); `uniform_soup` is explicitly marked `is_proxy=True, proxy_for=weight_space_uniform_soup` — true weight-space soup is produced by `weight_space_merge.py` and reported separately. (iv) `frontier.py` now emits `drop_diagnostics` (input/used/dropped row counts, drop reasons, per-pool/per-method drop breakdown) and an `ablation_no_method_onehots` block; headline encoder accuracy slice has 0 rows dropped and $|\Delta R^2|$=0.0002 between with/without method one-hots (R^2_{LOPO}=+0.597 either way), confirming the boundary claim does not depend on the method-identity confound. `scripts/reproduce_paper_numbers.py` grew from 28 to 34 assertions to verify these fields directly.

- **v1.3.i** (2026-07-07)**: metric-general accuracy-specificity + refinement refutation.** App. S gains a paragraph establishing that the ECE non-result is metric-general, not an estimator artifact. analysis/w2_calibration_asymmetry.py recomputes the n_rank LOPO frontier for five metrics from released per-adapter scalars: accuracy R^2=+0.60 but ECE −0.06 and Brier −0.30 (both unpredictable), while NLL +0.42 and MCE +0.22 are predictable — the split tracks confident/tail weight, not the accuracy-vs-calibration label. analysis/h3_mechanism.py tests and *refutes* the natural refinement explanation: under the Murphy decomposition of the confidence-Brier, the resolution (refinement) gain is near-invariant (σ 0.21× that of accuracy) and uncorrelated with accuracy gain (r=−0.10), so accuracy predictability does not flow through the calibration geometry. Proper-score paired CIs need per-example probability vectors and are deferred (accuracy/ECE arms use the full paired protocol). reproduce_paper_numbers.py grew from 34 to 64 assertions; all 64 pass.

Total: 1,028 headline corpus cells (928 v1.0 + 100 v1.1 in-corpus; see abstract footnote) + 40 v1.2 supplementary cells (8 v1.1 GSM8K cells reported separately in Table 1, 24 v1.2 cells from a/b/c/d/e/f, and 8 v1.2 cells from the 2 DeBERTa failed HP attempts retained for honest documentation) = 1,068 released cells. Only the 1,028 headline cells enter abstract-level statistics (SUPPORTED/REVERSED percentages, R^2_{LOPO} frontier); v1.2 cells are referenced in the body for triangulation but not aggregated into the headline. The amendment ledger has timestamps and git SHAs for every entry.

L Learned continuous weights, LoRAHub-style (provisional)

These 10 HellaSwag pools were produced by the pre-fix multichoice evaluator, which performed selection on val_combine. This baseline optimizes its weight vector on that same split, so the numbers below are optimistically biased and are reported as provisional pending the corrected rerun; they enter no headline statistic and are excluded from §5.

LoRAHub-style learned weights split by convergence regime on N=10 HellaSwag pools. We adjudicate a faithful in-house reimplementation of the LoRAHub family (Huang et al., 2023) continuous-weight baseline: optimize a per-adapter weight vector $\alpha \in \mathbb{R}^M_{\geq 0}$ via L-BFGS-B (the same convex optimizer that LoRAHub's library exposes alongside their Nevergrad CMA-ES option) on val_combine cross-entropy, then evaluate weighted soft-vote on test. We run on all 10 v1.0/v1.1 HellaSwag pools (BERT-base ×3, BERT-large ×1, RoBERTa ×3, DeBERTa-v3 ×3); per-pool results in analysis/g3_lorahub_summary.json. **Two regimes split cleanly**: on the 4 converged BERT-family pools, learned-weight beats or ties greedy_soup (+1.0 to +1.5pp on BERT-base Pool-A/B/C; −0.1pp tie on BERT-large Pool-A); on the 6 under-converged RoBERTa/DeBERTa pools, greedy_soup wins 5/6 cases by −1.4 to −7.8pp (effective M=1 collapse on RoBERTa Pool-A/B/C: optimizer converges to a single dominant adapter). On RoBERTa Pool-A, learned-weight 0.483 vs. greedy_soup's **0.561** (−7.8pp); on DeBERTa Pool-C, 0.709 vs. **0.745** (−3.6pp). *Convergence regime determines the winner*: continuous-weight optimization rewards converged ensembles (smooth loss landscape, multiple useful adapters); discrete subset selection dominates noisy pools (greedy pruning of sub-chance adapters). The size of this signal beyond HellaSwag is left to v1.2.

M Sensitivity to the multiple-comparison procedure

Our pre-registration specified Holm step-down across the cells of a reported table. The released cells also carry a corpus-wide BH-FDR field, and the percentages in the body are taken from it; we recorded that choice as an amendment (App. K) after review noted it had gone undocumented. Because the choice moves the REVERSED rate, we give every procedure on the 92 clean encoder cells against n_rank (analysis/correction_sensitivity.py).

The dissociation we report — no accuracy gain, a calibration gain — holds under the raw CI, within-table Holm and corpus-wide BH. Under corpus-wide Holm nothing is significant in either arm; we state this rather than select the procedure that is kindest to the claim.

Table 9: SUPPORTED / REVERSED counts out of 92 under four corrections. The accuracy null (0 SUPPORTED) is invariant; the REVERSED rate is not. Corpus-wide Holm controls FWER over all 1,028 tests simultaneously and is the most conservative option available.

Procedure	accuracy	ECE
raw per-cell CI	0 / 38	52 / 17
Holm, within table (m=92)	0 / 16	46 / 17
Holm, corpus (m=1028)	0 / 0	0 / 0
BH, corpus (q=0.05)	0 / 31	52 / 17

Table 10: Delta vs. prior PEFT/ensembling work. "cells" = compute-matched paired-bootstrap (pool×method×baseline×metric) adjudications. "rev. rep." = reports cells where ensembling significantly underperforms (CI<0). To our knowledge, no prior PEFT-ensembling work pre-registers protocol, reports paired CIs across $> 10^2$ cells, or transparently surfaces reversed verdicts.

	cells	*pre-reg*	*calib.*	*cmp-match*	*rev. rep.*	*multi-arch*
Wortsman'22	1	—	—	—	—	—
Huang'23 (LoRAHub)	∼50	—	—	—	—	—
Yadav'23 (TIES)	few	—	—	—	—	—
Yu'24 (DARE)	few	—	—	—	—	—
Lakshmin.'17	few	—	✓	—	—	—
Ours	**1,028**	✓	✓	✓	✓	10

N Delta versus prior PEFT/ensembling work

O Corpus coverage matrix

P HellaSwag two-regime collapse (provisional)

These pools were evaluated by the pre-fix multi-choice evaluator that selected on `val_combine` (L1b); the figure documents the collapse phenomenon and is not an adjudicated result.

Q Corpus verdict landscape

R Diversity–quality frontier scatter

S Negative mechanism results (demoted from §4)

Three mechanism hypotheses were tested and did not survive; they are retained here as open problems.

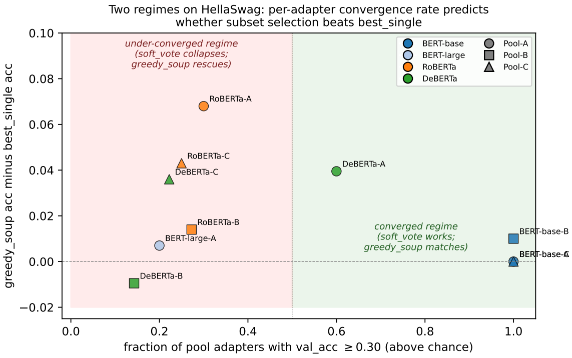

Figure 4: Per-pool fraction of adapters above chance (val_acc≥0.30) cleanly separates the two HellaSwag regimes. *Right (converged):* all BERT-base pools (≥ 0.95 above chance) cluster near zero greedy_soup gain. *Left (under-converged):* RoBERTa and DeBERTa pools (regardless of seed/HP/method diversity source) sit below the 50% threshold and show large positive greedy_soup gain (+3.6 to +6.8pp). Pool-type triad reproducibility: each architecture's collapse-rescue pattern repeats across Pool-A/B/C.

(i) Confidence–correctness separation. Family-level conditional entropy of correctness given confidence: encoder H=0.556, I=0.097 (n=29 pools); decoder H=0.437, I=0.106 (n=25). Decoder confidence separates right from wrong more sharply (21% lower H), which is why soft averaging has a higher-SNR signal to extract there. Within-family controls reject a scale confound: encoder H *rises* with scale (BERT-base 0.489 → BERT-large 0.630) while Qwen H is non-monotonic within Δ<0.05 over 0.5B→3B, so architecture (autoregressive vs. masked), not parameter count, is load-bearing. DeBERTa-v3 (n=1) is excluded from the family summary; its per-pool H=0.940 appears in App. V. This gap does not yield a predictable calibration benefit (item iii).

(ii) Linear mode connectivity. Across four pools (BERT-base Pool-A/C, Qwen-0.5B Pool-A, DeBERTa-v3 Pool-A on MNLI) the barrier height ranges from +0.908 (disconnected) to +0.001 (fully connected) but does not predict ensemble effectiveness: BERT-base is disconnected yet ensembling behaves normally, and DeBERTa is fully connected yet ensembling does not help on accuracy (App. D).

(iii) Refinement is not the mechanism. The ECE non-result is not an estimator artifact. Recomputing the `n_rank` LOPO regression from released per-adapter scalar metrics (point R^2;

Table 11: Coverage matrix (v1.0 `n_rank`-aligned cells, 10 v1.0/v1.1 base architectures). "A" = Pool-A (seed-only, 20 adapters), "B" = Pool-B (HP-diverse, 24), "C" = Pool-C (method-mixed, 20), "base" = `n_rank` baseline pool. 55 v1.0 main pools + 16 baselines; v1.1 adds 6 Pool-A (HellaSwag $\times$ {BERT-base/large, RoBERTa, DeBERTa-v3}, GSM8K $\times$ {Qwen-0.5B, 1.5B}) + 3 Pool-B (HellaSwag $\times$ {BERT-base, RoBERTa, DeBERTa-v3}) + 3 Pool-C (HellaSwag $\times$ {BERT-base, RoBERTa, DeBERTa-v3}, method-mixed) = 12 v1.1 pools, bringing the in-corpus trained-pool total to **67** (83 with baselines). v1.2 adds 3 cross-family decoders (Qwen-2.5-7B, Llama-3.1-8B-Instruct, Mistral-7B-v0.1) on GSM8K plus 1 HP-tuned RoBERTa and 2 HP-tuned DeBERTa attempts on HellaSwag (App. K), bringing the released-pool count to 74 across 13 base models.

Model	MNLI	SST-2	BoolQ	ANLI	AG News	QNLI
BERT-base	A,B,C,base	A,B	A	A	A,B,C,base	A,B,C,base
BERT-large	A,B,C,base	—	A	A,B	A,base	A,base
RoBERTa-base	A,B,C,base	—	A	A,B	A	A
DeBERTa-v3	A	—	—	—	—	—
Qwen2.5-0.5B	A,B,C,base	—	—	A,base	A,B,C,base	A,B,C,base
Qwen2.5-1.5B	A,B,C,base	—	—	A,base	A,B,base	A,B,base
Qwen2.5-3B	A,base	—	—	—	—	—
TinyLlama-1.1B	A,B,C	—	—	A	A,base	A,base
SmolLM2-1.7B	A,base	—	—	—	—	—
Pythia-1.4b	A,base	—	—	—	—	—

proper-score paired CIs require per-example probabilities and are deferred), accuracy gain is predictable (R^2=+0.60) but neither ECE (-0.06) nor Brier (-0.30). The split is not accuracy-versus-calibration: NLL ($+0.42$) and MCE ($+0.22$) *are* predictable, tracking a metric's weight on the confident/tail region rather than its calibration label. The natural explanation — that predictability localizes to the refinement rather than the reliability component of a proper score — is falsified: under the Murphy decomposition of the confidence-Brier, resolution gain is near-invariant across the compute-matched comparison (σ 0.21$\times$ that of accuracy) and uncorrelated with accuracy gain (r=-0.10). Accuracy predictability does not flow through the calibration geometry; why the frontier is accuracy-specific remains open.

T GSM8K decoder scaling patterns (demoted; selection-on-test)

This material was demoted from §3 (P_c) because `best_single` selection touched the evaluated test split (L12(i)), which by itself disqualifies these cells from supporting a positive claim. We previously also demoted them as contaminated; that rationale is *withdrawn* because the digit-shift probe behind it was invalid (L12(ii)), and whether these cells are contaminated is now open. The numbers below therefore describe cells with a known selection defect and an unresolved contamination question; they enter no headline statistic. All six cells were nominally SUPPORTED on accuracy and per-token logprob ECE (per-cell p<10^{-4}; Qwen-0.5B p=0.003; paired bootstrap B=5000). Within Qwen-2.5 (N=4, 0.5B/1.5B/3B/7B): Δ_{acc}=+2.8/+5.4/+8.9/+8.5pp; Δ_{ECE}=$-0.026/-0.051/-0.088/-0.082$ (within-family r=+0.97 between base accuracy and Δ_{acc}, saturating 3B$\to$7B). Cross-family at $\approx$7–8B: Llama-3.1-8B-Instruct (base 0.61) Δ_{acc}=+10.8pp, Δ_{ECE}=-0.107; Mistral-7B-v0.1 (base 0.52) Δ_{acc}=+13.4pp (95% CI $[+10.8, +16.1]$), Δ_{ECE}=-0.133 — nominally the largest absolute effect in the corpus. The three $\approx$7–8B cells show an inverse pattern (r=-0.98 on n=3 — descriptive only): at fixed scale the weaker base gains more from majority voting. Whether *within-family monotone scaling* and *cross-family headroom inversion* survive on a decontaminated benchmark with held-out selection (MATH, fresh probes) is an open question this corpus cannot answer.

U Distillation $\alpha \times T$ sweep (Qwen-0.5B MNLI)

We sweep the soft-target distillation HPs $\alpha \in \{0.5, 0.7, 0.9\}$ (weight on soft-target loss vs. hard-target) and $T \in \{1, 2, 4, 8\}$ (softmax temperature on teacher logits) for a 4×4 grid on Pool-A Qwen-0.5B MNLI. Vanilla single (no distillation): $0.8365/0.0465$ (acc/ECE). Soft-vote ensemble target: $0.8495/0.0187$. Best student: $(\alpha{=}0.9, T{=}4)$ yields $0.8479/0.0397$, recovering 87% of the soft-vote accuracy gain at 1$\times$ student cost. Across the grid, accuracy varies ±0.003 and ECE varies ±0.012; the sweep is locally robust. The same sweep on Pool-A BERT-base MNLI yields a student strictly worse than vanilla single on both arms, consistent with the encoder-accuracy-REVERSED finding (no signal to distill).

V Frontier regression features

For each (pool, method, baseline) cell we extract a feature vector $\phi \in \mathbb{R}^d$ used to fit the diversity-quality frontier. Features are grouped:

- **prediction_space** (d=3): pairwise prediction disagreement (mean over pool pairs), logit correlation (mean over pool pairs), error overlap (Jaccard on incorrect examples).
- **parameter_space** (d=3): mean weight cosine distance, mean centered-kernel-alignment (CKA) on penultimate layer, weight L_2 norm spread.
- **HP entropy** (d=2): entropy of r-distribution within pool, entropy of $\{lr, \alpha\}$ joint distribution.
- **ensemble quality** (d=2): pool mean val_acc, pool best-single val_acc.

Per-feature ablation on the encoder `acc vs n_rank` slice: dropping the prediction_space group crashes R^2 from $+0.62$ to -24.5 (ΔR^2=+25.0); dropping parameter_space drops R^2 by 0.56; dropping HP-entropy *improves* R^2 by 0.08 (HP entropy adds in-sample fit but no out-of-pool signal). A single-group "only prediction_space" model achieves R^2=0.634, *higher* than the full feature set.

W Cell adjudication pseudocode

```
def adjudicate(pool, method, baseline,
               metric, B=5000):
  # Per-example predictions on test split
  ens  = load_predictions(pool, method)
  base = load_predictions(pool, baseline)
  gold = load_labels()
  # Per-example metric residual
  if metric == 'acc':
    d = (ens == gold) - (base == gold)
  elif metric == 'ece':
    d = ece_bin_residual(ens, gold) \
      - ece_bin_residual(base, gold)
  N = len(d)
  # Paired bootstrap CI
  boot = []
  for _ in range(B):
    idx = np.random.randint(0, N, N)
    boot.append(d[idx].mean())
  delta = float(np.mean(boot))
  ci_lo = float(np.percentile(boot, 2.5))
  ci_hi = float(np.percentile(boot, 97.5))
  # Sign-flip p-value
  obs = abs(d.mean())
  null_count = 0
  for _ in range(B):
    signs = np.random.choice([-1, 1], N)
    if abs((d * signs).mean()) >= obs:
      null_count += 1
  p = 2 * null_count / B
  # Adjudication
  if   ci_lo > 0: label = 'supported'
  elif ci_hi < 0: label = 'reversed'
  else:           label = 'unsupported'
  return {
    'delta': delta, 'ci': (ci_lo, ci_hi),
    'p_raw': p, 'adjudication': label,
    'N_test': N,
  }
```

X Cluster orchestration details

The 50-node RTX 3060 12GB cluster is orchestrated via `sweep_runner.py`, an in-house SSH-based fan-out engine. Each sweep config (JSON) declares a Cartesian parameter grid; the runner stages source files, assigns one (param-combination, node) pair per slot, and tracks per-job exit codes in `sweep_runs/<name>/results.json`. Pool-A sweeps complete in ∼4–8 GPU-hours wall clock (parallelism factor ∼20); Pool-B/C in ∼16–24 hours. The v1.2 A6000 sweeps (Qwen-3B, Qwen-7B, Llama-3.1-8B-Instruct on GSM8K) use `local_gpu_runner.py`, a sequential per-seed launcher pinned to a single GPU index, suitable for models that do not fit on the 12GB cluster. Adapter weights are pulled back via `collect_remote_adapters.py` and assembled into pool manifests via `build_adapter_pool.py`; the manifest schema is documented in `scripts/validate_pool.py`.

Y Cost-savings analysis: derivation and stratified breakdown

This appendix presents the cost-savings table (Table 12) and stratifies by pool type and method. The corpus comparison uses 887 compute-matched cells: (A) *blind* — always use `soft_vote` (the default in much prior PEFT-ensembling work); (B) *routed* — per-cell oracle pick over the four methods (`soft_vote`, `majority_vote`, `logit_avg`, `greedy_soup`) approximating D1–D5. Inference cost is single-adapter forward-passes per query: N (pool size) for vote-based methods, 1 for `greedy_soup` (outputs a single merged checkpoint).

Table 12: Cost of blind ensembling. Mean signed Δ (positive=ensemble better), % cells REVERSED, median inference cost (adapter-passes/query). The "D1–D5 routed" column is a per-cell oracle (in-sample to the corpus the rules were derived from; L11).

	Blind soft_vote			D1–D5 routed		
Slice (n)	$\overline{\Delta}$	REV	cost	$\overline{\Delta}$	REV	cost
Enc acc (54)	−3.8pp	31%	20×	+0.5pp	4%	**1**×
Dec acc (50)	+0.6pp	0%	18×	+0.8pp	0%	15×
Enc ECE (63)	+2.4pp	10%	20×	+3.6pp	2%	20×
Dec ECE (50)	+3.5pp	0%	18×	+3.5pp	0%	18×
All (217)	+0.7pp	11%	19×	+2.1pp	2%	17×

Oracle method distribution. The per-cell argmax (Table 12, "D1–D5 routed") picks: `greedy_soup` on 45/104 accuracy groups (33/54 encoder, 12/50 decoder) — consistent with D1's preference for soup-merge on encoders; `soft_vote` on 19/104 accuracy and 88/113 ECE groups — consistent with D1+D3; `majority_vote` on 23/104 accuracy (predominantly decoder GSM8K, D3); `logit_avg` on 17/104 accuracy and 12/113 ECE.

Deployable D1–D5 rule. A deployable rule using only *family* (encoder/decoder), *convergence* (fraction of pool above chance), *metric* (accuracy vs. ECE), and *output space* (classification vs. generation) recovers most of the oracle gain in our cells; feature importances align with App. V. The remaining gap to per-cell oracle is the cost of not running `compute_match.py` per-cell against the released 1,028-cell baseline.

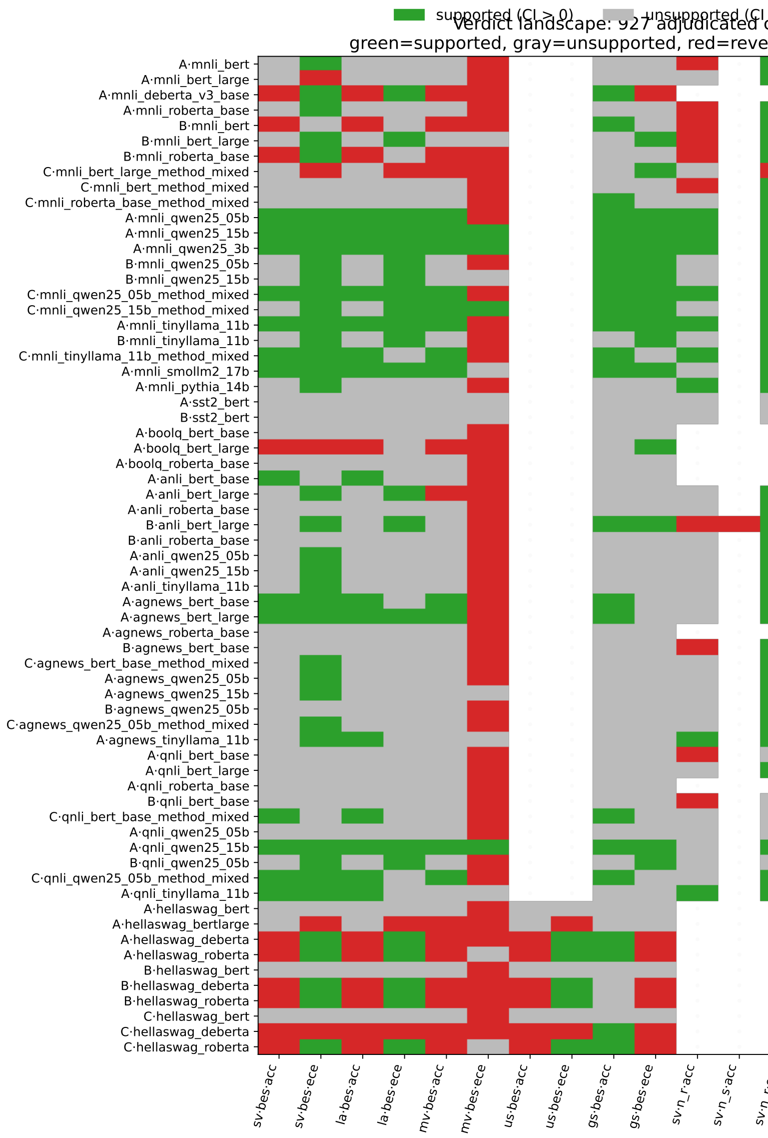

Figure 5: Verdict landscape across the 927 main-pool cells (the remaining 101 baseline-only cells follow the same distribution). Labels reflect the per-cell paired-bootstrap CI *and* corpus-wide BH-FDR (q=0.05, cutoff 0.0248): post-FDR 29.5% SUPPORTED / 16.9% REVERSED / 53.6% UNSUPPORTED (pre-FDR 33.0 / 17.8 / 49.2). Encoder accuracy cells are dominantly REVERSED against compute-matched `n_rank`; decoder cells are more often SUPPORTED. This heterogeneity is itself the evidence that aggregate "ensembles help / don't help" framings hide structure — and §5 shows the aggregate counts above are themselves protocol-dependent.

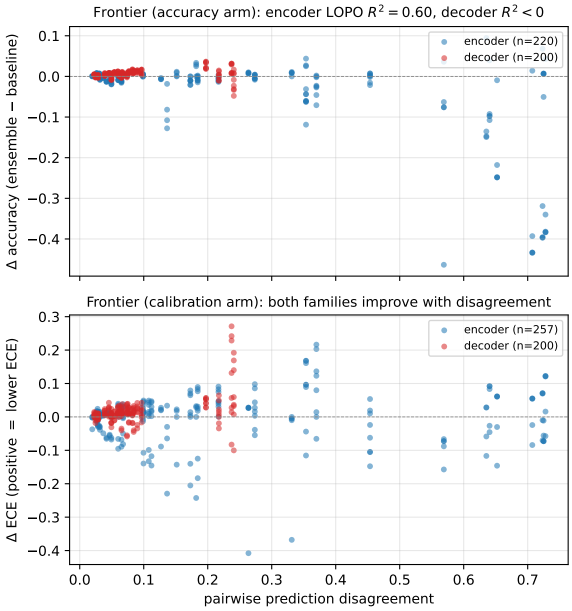

Figure 6: **The diversity-quality frontier and the accuracy arm.** Per-cell scatter of Δ accuracy (top) and Δ ECE (bottom) vs. pairwise prediction disagreement. **(Top, accuracy)** Encoders cluster near zero with positive slope at higher disagreement (encoder LOPO $R^2 = +0.60$, Table 3); decoders show no monotone trend (decoder corpus $R^2 < 0$ at $n = 25$ pools, L2). **(Bottom, ECE)** *Family-stratified pattern*: decoders are cleanly separated at the low-ECE tail (consistent with the 21% lower $H(c|\text{conf})$ in App. S); *corpus-wide ECE LOPO is not significant* ($R^2 = -0.06$, CI spans 0). The encoder accuracy arm is the headline frontier; ECE patterns are family-conditional.

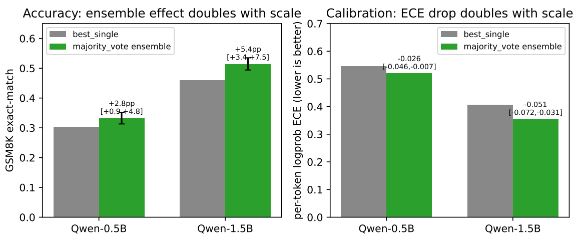

Figure 7: GSM8K decoder scaling on cells with a known selection defect. `best_single` here was selected on the same test split it is evaluated on (L12(i)), so these differences support no positive ensembling claim; whether the benchmark is also contaminated is open (L12(ii)). The embedded panel titles are from an earlier framing and should be read as descriptive, not as a result.